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Respiratory tract infections: an update on the complexity of bacterial diversity, therapeutic interventions and breakthroughs

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Abstract

Respiratory tract infections (RTIs) have a significant impact on global health, especially among children and the elderly. The key bacterial pathogens *Streptococcus pneumoniae*, *Haemophilus influenzae*, *Klebsiella pneumoniae*, *Staphylococcus aureus* and non-fermenting Gram Negative bacteria such as *Acinetobacter baumannii* and *Pseudomonas aeruginosa* are most commonly associated with RTIs. These bacterial pathogens have evolved a diverse array of resistance mechanisms through horizontal gene transfer, often mediated by mobile genetic elements and environmental acquisition. Treatment failures are primarily due to antimicrobial resistance and inadequate bacterial engagement, which necessitates the development of alternative treatment strategies. To overcome this, our review mainly focuses on different virulence mechanisms and their resulting pathogenicity, highlighting different therapeutic interventions to combat resistance. To prevent the antimicrobial resistance crisis, we also focused on leveraging the application of artificial intelligence and machine learning to manage RTIs. Integrative approaches combining mechanistic insights are crucial for addressing the global challenge of antimicrobial resistance in respiratory infections.

Keywords Respiratory tract infections · Antimicrobial resistance · Therapeutics · Genomics · Systems biology · Structural bioinformatics

Introduction

Respiratory tract infections (RTIs) are a major cause contributing to the high mortality and morbidity rates worldwide. These range from asymptomatic or mild to severe infections resulting in terminal conditions (Calderaro et al. 2022). Most of the RTI cases are observed in young children (aged < 5 years) and in elderly people (aged > 65 years). Several pathogenic bacterial and viral agents account for the leading cause of RTIs-associated cases. These pathogenic bacteria have a common entry point through respiratory mucosa includes epithelium, which is supported by basal lamina and lamina propria, together constituting the extracellular matrix (ECM) providing structural support. The receptors expressed by the ECM represent adhesive surfaces for bacterial adhesins, facilitating contact between the host epithelial cells and pathogenic bacteria (Sarshar et al. 2021). This interaction promotes bacterial colonization, subsequently triggering inflammation in the respiratory tract.

RTIs can be categorized into two main types based on their focus of infection, namely, upper respiratory infections

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(URTIs) and lower respiratory infections (LRTIs). The infections associated with URTIs are mostly mild and results in rhinitis, sinusitis, otitis, pharyngitis and epiglottitis. However, LRTIs are observed to cause more severe ailments such as bronchitis, emphysema and pneumonia. Compared to URTIs, LTRIs often cause complicated infections, leading to a more fatal outcome (Saleri and Ryan 2019). The common symptoms associated with LRTIs include weakness, fatigue, shortness of breath and coughing that last from around one to three weeks (Alam and Bastakoti 2015). According to the World Health Organisation (WHO), respiratory infectious diseases occupy first place in the ranking for the burden of the disease reported by years of life lost through death and disability (Avenidaño Carvajal and Perret Pérez 2020). According to the Global Burden of Diseases (GBD) studies conducted in 2016, 336 million cases of pneumonia were from low-middle-income countries and among them, 62% (~2 million) deaths turned out to be in sub-Saharan Africa and South Asia (GBD 2016 Lower Respiratory Infections Collaborators 2018). The number of deaths reported by acute respiratory tract infection in children was estimated to be around 1.9 million, of which 20% of them were reported in India (Chandy et al. 2022). The latest forecast report proposed by the GBD for the period 2022–2050, respiratory infection is projected to account for 46.7% of all infections in 2022, with this proportion expected to rise up to 76.41% by 2050. Specifically, the incidence of URTIs is anticipated to double by 2050 (Vollset et al. 2024). LRTIs are usually treated with antibiotics, but in some cases, the disease is beyond treatable and becomes life-threatening (Weidmann et al. 2022).

Microbial pan-genome

The microbial pan-genome is the complete set of genes that are shared by all the strains within a species. It encompasses two distinct components: the core genome and the accessory genome. The core genome comprises genes that are commonly found in all or nearly all strains and are usually those involved in essential functions such as replication, metabolism, and gene expression. (Karampatakis et al. 2024). While, accessory genome consists of more variable genes that are present in some but not all strains and may, therefore, include those conferring strain-specific adaptations like antibiotic resistance, virulence factors, or the ability to thrive in certain environments. These variable genes have the potential to be carried by mobile genetic elements (MGEs), including plasmids, transposons, and prophages, which promote horizontal gene transfer (HGT) between adaptive genes. *Acinetobacter baumannii* (*A. baumannii*) has many antibiotic-resistance genes in its accessory genome, which have the capability to disseminate to

different strains. A similar pattern is observed in *Klebsiella pneumoniae* (*K. pneumoniae*) strains, which have led to the emergence of antimicrobial resistance (AMR) as well as hypervirulent strains causing global public health concerns (Karampatakis et al. 2023). Core genes are often enriched for essential housekeeping functions and even more, they have unknown functions, highlighting the gaps in our understanding of bacterial diversity (Freschi et al. 2019).

Resistome

The resistome comprises a complex of resistant genes (indigenous and acquired) that confers tolerance to antimicrobial agents in bacteria. Resistance genes are transferred between species through HGT, enabling the spread of resistance traits. Three mechanisms of AMR exist such as persistence, resistance, and tolerance. Recent studies have shown that resistance is marked by increased minimal inhibitory concentration (MIC) in bacterial populations, whereas tolerance and persistence do not exhibit increased MIC (Algammal et al. 2023) to commonly used antimicrobial agents. The prevalence of virulence genes in aquatic environment is similar to that in clinical isolates, suggesting this environment may be a potential source of infections due to organisms such as *Pseudomonas aeruginosa* (*P. aeruginosa*). Multidrug resistance (MDR) in *P. aeruginosa* results from the interplay between intrinsic and acquired resistance mechanisms, affecting various antibiotic groups such as anti-pseudomonal, β -lactams, fluoroquinolones, aminoglycosides and polymyxins (Behzadi et al. 2022a). The spread of β -lactamases, such as Metallo-beta-lactamases (MBLs), is a significant concern, as they neutralize β -lactam antibiotics through hydrolysis (Behzadi et al. 2020).

Virulome

The virulome is a collection of genes that contributes virulence to opportunistic pathogenic bacteria. These bacteria possess different virulence genes capable of infecting the lower airways and lung parenchyma (Peek et al. 2018). The recent pan-genomic studies have revealed that MGEs such as transposons, integrons and bacteriophages play a crucial role in transmitting virulence factors and antimicrobial resistance genes (ARGs) between bacteria. MGEs-carrying ARGs are often carried by plasmids and can be transferred through various types of HGT, including phage-mediated HGT, which has been shown to transfer chromosomal ARGs between *A. baumannii* strains (Wachino et al. 2019; Karampatakis et al. 2023). In addition, the virulence in *A. baumannii* is attributed to different factors such as capsular exopolysaccharide, efflux pumps, lipopolysaccharide, lipooligosaccharide, outer membrane proteins, pili metal ion

uptake systems, two-component systems and secretion systems (Karampatakis et al. 2024). Recent studies have shown that hypervirulent *K. pneumoniae* strains are largely composed of antibiotic-sensitive populations, with the exception of ampicillin. Conversely, convergent *K. pneumoniae* strains have been identified as MDR strains that carry the aerobactin synthesis locus (*iuc*) and produce carbapenemase enzymes (Ahmadi et al. 2022; Karampatakis et al. 2023).

RTIs and bacterial causative agents

The management of RTIs becomes complicated due to the inaccuracy in identifying the common etiological factors and administration of appropriate antibiotics. The common microbiological causative agents associated with RTIs include pathogenic bacteria such as *Streptococcus pneumoniae* (*S. pneumoniae*), *Haemophilus influenzae* (*H. influenzae*), *Staphylococcus aureus* (*S. aureus*), *Streptococcus pyogenes* (*S. pyogenes*), *Enterococcus* spp, *Klebsiella pneumoniae* (*K. pneumoniae*), *Pseudomonas aeruginosa* (*P. aeruginosa*), *Acinetobacter baumannii* (*A. baumannii*) and *Mycobacterium* spp. (Ravi et al. 2023). These microbes colonize and infect various parts of the respiratory tract and prolong the infection using multiple virulence mechanisms

(Fig. 1). Among Gram-positive bacteria *S. pneumoniae* involves various virulence factors such as *PLY*, *IgA1* protease, pili, and *PavaA* (Aceil and Avci 2022); while, *S. pyogenes* involves *M* proteins, streptolysin S & O, *SpeB*, and *Sags* (Fontaine et al. 2003) *S. aureus* involves *PVL*, *PSMs*, and *TSST1* (Tam and Torres 2019). In the case of *Mycobacterium* spp. virulence factors such as *Smurf1*, *T7S*, and *Mce/ZmpL/PtpA* (Netea et al. 2006) are involved; Among Gram-negatives, *P. aeruginosa* involves *Pili*, *Chop*, and *TNFα* (Curran et al. 2018); *A. baumannii* involves *ExoS/ExoY*, and *OMPs* (Zhang et al. 2022); *K. pneumoniae* involves *LPS*, *Pili*, and *Siderophore* (Abbas et al. 2024); *H. influenzae* involves *LPS*, *Pili*, *Hia*, and *IgA1* protease (King 2012). These bacterial pathogens serve as key players in respiratory infections that can be life-threatening, especially in individuals with immuno-compromised conditions, infants and young children, or other risk factors. Pathogenic bacteria involved in RTIs further discussed as follows:

Streptococcus pneumoniae

The facultative anaerobic *S. pneumoniae* are Gram-positive cocci that appear usually in pairs. As reported by Dion and his team, about 20 – 30% of healthy adults and 40 – 50% of

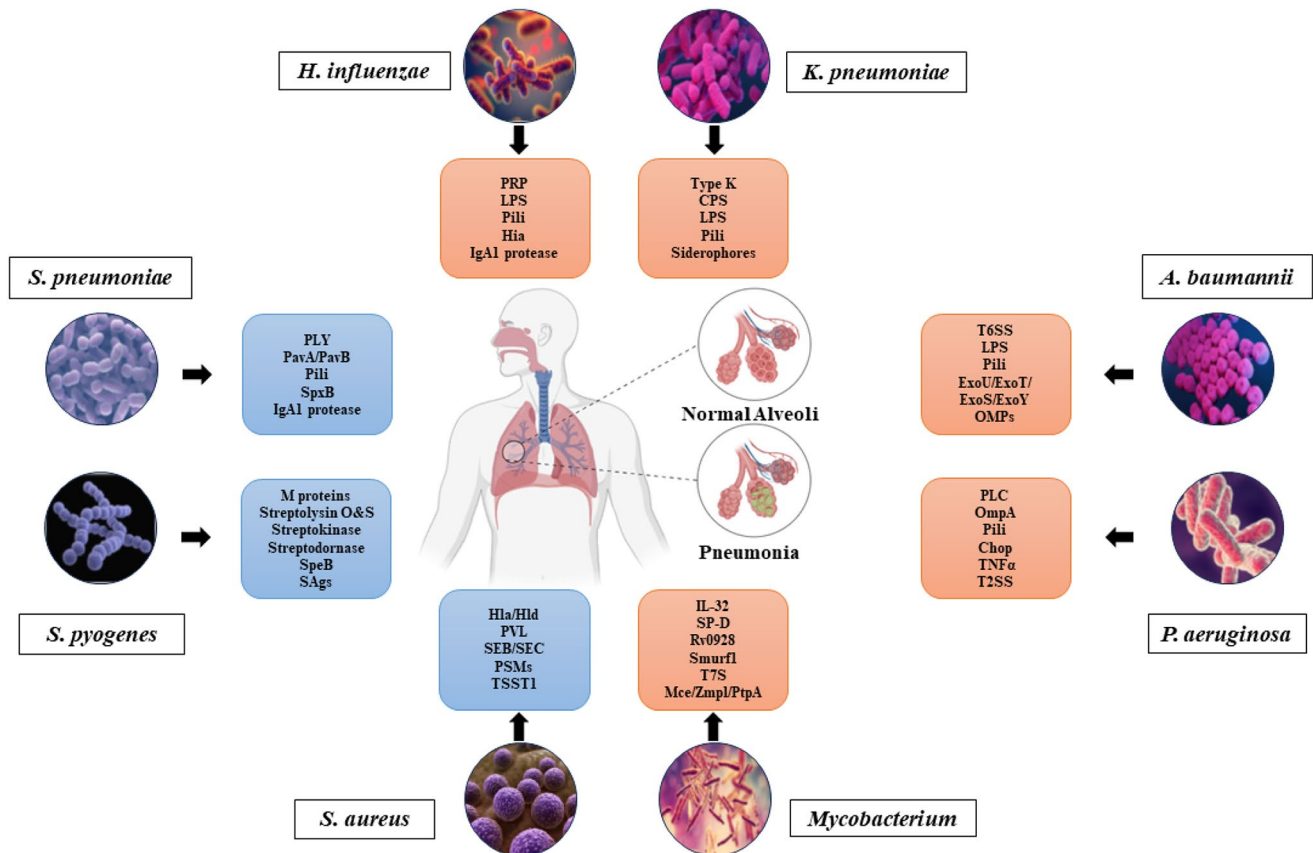


Fig. 1 Key underlying molecular virulence mechanisms involved in RTIs

healthy children are carriers (Dion and Ashurst 2024). The bacterial infections are often observed among children < 5 years and adults ≥ 65 years (Manoharan et al. 2017). The common route of transmission is by aerosols, secreted saliva, or direct contact from the infected patients, which results in the colonisation of the bacteria in the nasopharynx region. Colonisation in the nasopharynx region gain access to the upper respiratory tract, where it binds to the epithelial cells initiating the infection (Mekuria et al. 2022). The bacteria take over the host system by targeting the plgR (polymeric immunoglobulin receptor) present on the apical side of the host cells. This gets carried in the form of vesicles through a process known as endocytosis, which is then translocated to the basolateral surface which is more exposed to the inner lining of the host tissue. The development of pneumonia is dependent on the ability of *S. pneumoniae* to establish a LRTIs, even in the presence of the host defense mechanisms aspirated in the lungs (Loughran et al. 2019; Peela et al. 2023). The pathogenicity is, through the virulence factors, crucial for its ability to cause disease, allowing the bacteria to evade the host's immune system (Kadioglu et al. 2008). One of the key virulence factors is the polysaccharide capsule, which allows the bacteria to escape the nasal mucus and adhere it to the epithelial cells in the nasopharynx. Another virulence factor pneumolysin, a cytoplasmic toxic potentiates to forms pores in host cells ultimately facilitating the transmission of the bacteria (Subramanian et al. 2019). Pneumococcal surface protein (PspA and PspC) helps in colonization by binding to the epithelial cell membrane, which is anchored to the cell wall via phosphorylcholine (PCho) (Du et al. 2021). Other choline-binding proteins like LytB, LytC, CbpC, and CbpC modify proteins on the cell surface, contributing to host cell identification and thereby facilitating binding to host cell receptors. Pili promotes adherence and colonisation within the nasopharynx, along with preventing them from being phagocytosed by the immune cells. Neuramidase cleaves the salic acid molecule, thereby degrading the mucus layer and promoting bacterial growth survival. IgA1 protease breaks down IgA antibodies in the extracellular environment, further aiding the bacterial survival (Brooks and Mias 2018).

To combat the pathogenicity of *S. pneumoniae*, different treatment strategies have been employed, including the use of vaccines, which are classified into two types that are whole cell vaccines and subunit vaccines. The whole-cell vaccines being either weakened or inactivated vaccines, while subunit vaccines such as polysaccharide vaccines, conjugate vaccines and protein-based vaccines (Kim et al. 2017). According to Centres for Disease Control and Prevention (CDC), the commercially available vaccines include pneumococcal conjugate vaccines (PCVs) such as PCV10 and PCV13, which are recommended for children

under 5 years old and pneumococcal polysaccharide vaccine (PPSV23), in individuals between 5 and 64 years old who are at higher risk and 65 years and older. These vaccines are important in protection against pneumococcal disease, and understanding their different types vaccine is essential for effective prevention and treatment of diseases.

Haemophilus influenzae

H. influenzae, causes infections such as epiglottitis, sinusitis, and otitis media, specifically in children, elders, and immunocompromised patients (Tsang and Ulanova 2017). The transmission of *H. influenzae* occurs through the inhalation of bacteria-containing cough droplets or by respiratory secretion through direct contact. Colonization occurs when the patient is immunocompromised, which may lead to pneumonia. In contrast with other bacteria, *H. influenzae* does not produce any toxins. However, the inability of the immune targets certainly damages the surrounding tissue, causing bacterial invasion and colonization. Once the adherence to the host is completed, *H. influenzae* invades the cells in the respiratory tract and immune system (Khattak and Anjum 2023). This Gram-negative pleomorphic bacilli possess a capsule consisting of a polymer of ribose and ribitol-5-phosphate (polyribosylribitol – PRP) that is responsible for the high colonization capacity of the bacteria in the upper respiratory tract. The capsule contributes as the major cause of infections in many parts of our body (Wang et al. 2021). However, not all serotypes of *H. influenzae* have capsules. Around six different types of serotypes have been reported that have been designated as “a-f,” which have specific capsular polysaccharides. Among the capsulated (typeable) serotypes, the most virulent serotype is type b (Hib), which is commonly detected in infants (Slack et al. 2021). The *bexA* gene present in the *cap* locus encodes for the proteins that are involved in capsule transport in encapsulated *H. influenzae* (Fan et al. 2018). While non-capsulated *H. influenzae* (non-typeable) does not have a polysaccharide capsule, it makes them less invasive than capsulated *H. influenzae*. These Non-typeable *H. influenzae* (NTHi) are capable of causing chronic obstructive pulmonary disease (COPD). The outer membrane protein P6 acted as a potent molecular marker for the identification of NTHi, which was later taken over by other genes such as *iga*, *sodC*, and *hap* that were more prominently found in the NTHi (Chang et al. 2010). The development of Hib vaccines has significantly reduced the incidence of Hib-related diseases. The Hib polysaccharide conjugate vaccine is an inactivated bacterial vaccine that works by conjugating a protein carrier to polysaccharide capsule surrounding the Hib bacterium, which enhances immunogenicity, especially in young children. This gets activated by converting the polysaccharide from a

T-independent to a T-dependent antigen. There are five vaccine options available that includes three monovalent Hib polysaccharide-protein conjugate vaccines, ActHIB (PRP-T), Hiberix (PRP-T), and PedvaxHIB (PRP-OMP) and two combination vaccines that are Pentacel (DTaP-IPV/Hib) and Vaxelis (DTaP/Hib-HepB). These vaccines have been instrumental in protecting against Hib diseases (Vidor et al. 2001).

Staphylococcus aureus

S. aureus are Gram-positive cocci in clusters that are a common causative agent detected in RTIs. The bacteria are mostly detected in children < 5 years old and also in the adult > 70 years age group. The bacteria with pathogenic traits are prone to community spread, and it can be hospital-acquired (Paonessa et al. 2019). The most common route of transmission includes nose, skin and pharynx, leading to osteoarticular, pleuropulmonary and other infections. Aggravation of the infection leads to complicated lung infections that are more fatal than other infections (Tong et al. 2015; Pivard et al. 2021). The bacteria possess adhesive molecules such as clumping factor B (*ClfB*), iron surface determinant A (*IsdA*), and wall teichoic acid (WTA) that help in its interaction with the human cell components (Baur et al. 2014). *S. aureus* adheres and invades host epithelial cells using these molecules that are collectively termed MSCRAMM (Microbial surface components recognizing adhesive matrix molecules). Two adhesive molecules, *ClfB* and wall-associated teichoic acid factors, are known to be important for the adhesion and attachment to nasal epithelial cells of several bacteria (Liu 2009). The bacteria reach the nasal colonization zone through the hosts respiratory air intake. When it reaches the lungs, it adapts to the new environment and modifies the expression of factors compared to the nasal colonization state (Pivard et al. 2021). The initial exposure of *S. aureus* to host tissues beyond the mucosal surface triggers the upregulation of virulence genes. These virulence factors secrete toxins that play a key role in infections, which are mainly divided into three major groups such as pore-forming toxins (PFTs), exfoliative toxins (ETs), and superantigens (SAGs). The pore-forming toxins can be further divided into α -hemolysin (Hla), β -hemolysin, leukotoxins, and phenol-soluble modulins (PSMs). Hla helps in pulmonary epithelial disruption and invasion of the host defense. Similarly, β -hemolysin aids in epithelial phagolysosome formation, which facilitates escape from host defense. Phenol-soluble modulin α (PSM α) plays a role in biofilm dispersion within phagolysosomes, leading to host defense responses in epithelial cells and macrophages. These responses include the induction of necroptosis, neutrophil lysis, and the production of toxic shock syndrome

toxin 1 (TSST-1), which activates abnormal T-lymphocytes and triggers cytokine production in the lungs (Oliveira et al. 2018).

Over the last decade, *S. aureus* has been recognized as a clinically important intracellular pathogen due to its increasing infectious rates (Deinhardt-Emmer et al. 2019). The main public health concern caused by pulmonary *S. aureus* is critical nosocomial pneumonia, while *S. aureus* Community-Acquired Pneumonia (SA-CAP) accounts for only 5–6% of CAP cases as reported in the US and Europe (de Miguel-Díez et al. 2017). A detailed study done by Braverman et al. 2022; showed that *S. aureus* relies on a specific respiratory tract CD4⁺ Th1 cells which polarises memory cells that have the potential to target the cells in the vaccine that can prevent the growth of *S. aureus* causing pneumonia (Braverman et al. 2022). Developing a vaccine for *S. aureus* is particularly challenging as *S. aureus* is involved in the pathogenesis of many factors associated with its virulence. Previous research on vaccines focused on addressing key components of *S. aureus* disease, like immune evasion (capsules), host cell binding (ClfA), and nutrient acquisition (MntC). The Numerous Attempts to formulate vaccines against *S. aureus* vaccines have been categorized into two different methods that are passive immunization (transfer of *S. aureus* antigen-specific antibodies such as SEB, ClfA, and Altastaph) or active immunization (vaccination with recombinant antigens to induce protective antibody responses that are Glycovaxine, STEBvax, StaphVax, SA5Ag). Although, single antigen-based vaccines have proven to be useful to only a certain extent but have highlighted the importance of multi-antigen vaccination approaches (Scafa-Udriste et al. 2023).

Streptococcus pyogenes

S. pyogenes are Gram-positive cocci in chains, a major human-specific bacterial pathogen, can cause a wide range of symptoms, from minor localized infections to potentially fatal invasive infections. Based on the types of hemolysis on blood agar, they are divided into three groups: gamma hemolytic (no hemolysis), β -hemolytic (complete lysis of red blood cells) and α -hemolytic (green hemolysis). Group A streptococci (*S. pyogenes*) and group B streptococci (*S. agalactiae*) are the two types of β -hemolytic streptococci.

Infection caused by Group A streptococci (GAS) is the most common bacterial cause of pharyngitis, accounting for 15–30% of cases. Acute rheumatic fever (ARF) may result from untreated GAS pharyngitis in a small percentage of patients (0.3–3%). GAS, which has N-acetyl glucosamine (GlcNAc), is linked with rhamnose polymer. It is an important virulence determinant of GAS. The bacteria adhere and colonize the epithelial cells of the host mucosa in the upper

respiratory tract. Virulence factors of some *S. pyogenes* are encoded within chromosomes, and the remaining show properties of MGEs (Al-Bermani et al. 2020). According to the study by Alamiri et al., the M protein, capsule, SpeB, NADase are the most important virulence factors for the biofilm formation in GAS. Interestingly, the cysteine protease SpeB and toxins produced (such as SLS) also plays an inhibitory role in the uptake of planktonic GAS into the epithelial cells (Alamiri et al. 2020, 2023). The M protein is considered a crucial antigen of GAS vaccines; development in preclinical and clinical settings has shown in the last decade that M protein-based GAS vaccines have a greater potential than non-M protein-based vaccines. The clinical development of GAS vaccines have drawbacks, including the absence of defined immunity, insufficient animal models, variety of GAS strains and perceptions of risk of autoimmune complications triggered by the vaccine (Fan et al. 2024).

Klebsiella pneumoniae

K. pneumoniae is an opportunistic Gram-negative bacilli, causing pneumonia, particularly in people with conditions such as alcohol addictions or diabetes. This organism is known to colonize the oropharynx and intestinal mucosa of humans and becomes particularly dangerous once it enters the body due to its potentially high virulence and resistance to common antibiotics. *K. pneumoniae* infections are typically categorized as either occurring in the community or in healthcare facilities, with the latter being more commonly attributed to this microbe (Ashurst and Dawson 2024). Humans are the main carriers, with 1–6% carrying the bacteria in their nasopharynx, which can lead to hospital outbreaks through contact, particularly through transmission from the hands of healthcare workers (Miriti et al. 2023). While such infections are comparatively less common in Western culture, in other developing countries such as Africa, they are of great concern. Vulnerable individuals such as newborns, the elderly, and immunocompromised individuals are at higher risk (Bengoechea and Sa Pessoa 2019). The lungs innate defense system attempts to block such infections via the mucociliary clearance mechanism, which traps the pathogens in mucus and expels them. Nevertheless, *K. pneumoniae* has developed sophisticated mechanisms to invade and overcome the human immune system, especially by resisting ingestion and destruction by immune cells. Studies have identified the virulence factors, such as capsular polysaccharides, lipopolysaccharides and hair-like structures, namely, fimbriae, proteins on the outer membrane and siderophores, that contribute to the pathophysiology of *K. pneumoniae* (Abbas et al. 2024). Currently, there are no vaccines available against *K. pneumoniae* infection.

However, a 24-valent conjugate vaccine is in the pre-clinical stage, which combines the capsular polysaccharides (CPS) of *K. pneumoniae* and *K. oxytoca* with the O antigen of *P. aeruginosa*, using *Pseudomonas* toxin A as the carrier. Additionally, several vaccine candidates are under investigation, including formulations based on outer membrane proteins, Lipopolysaccharides, and CPS. These ongoing efforts aim to develop an effective vaccine against *K. pneumoniae* infection (Assoni et al. 2021).

Pseudomonas aeruginosa

P. aeruginosa is a Gram-negative ubiquitous and opportunistic bacilli. It weakens the immunity to adapt and evolve in the host organism (Garcia-Clemente et al. 2020). *P. aeruginosa* has a prevalence of 7.1–7.3% in RTIs compared to other healthcare-associated infections (Reynolds and Kollef 2021). Among the various infections caused by *P. aeruginosa*, pneumonia is the most commonly reported RTI (Khatkhat and Anjum 2023). Among them, nosocomial pneumonia is the most common. *P. aeruginosa* colonization progresses to LRTIs that can lead to chronic diseases such as asthma, COPD, bronchiectasis and cystic fibrosis (Garcia-Clemente et al. 2020). *P. aeruginosa* infection is difficult to treat due to its rapid mutational behaviour that adheres to resistance to antibiotics. They are mostly seen on medical devices that come in contact with wet surfaces/environments (Qin et al. 2022). A major adapting feature of the bacteria is to survive in hypertoxic environments or other harsh conditions by forming a biofilm. Alginate is an exopolysaccharide and is also considered as one of the main components of biofilm in *P. aeruginosa*. These alginates are attached to the mucin found in the respiratory tract that are adhesive in nature. The acetyl group present in the alginates increases the viscosity that in turn helps for the accumulation of nutrients and water in the biofilm (Tuon et al. 2022). *P. aeruginosa* uses its flagellum and pili to adhere and move through host tissue. The flagellar cap protein, FliD, plays a crucial role in this process by facilitating adhesion to mucins in the upper airways. On the apical surface of respiratory epithelial cells, flagella and pili bind to glycosphingolipids, specifically ganglio-*N*-tetraosylceramides. Additionally, on the basolateral surface, they bind to heparan sulfate, proteoglycans and N-glycans, promoting adherence and motility (Curran et al. 2018).

Multidrug-resistant (MDR) *P. aeruginosa* is more prevalent in terminally ill pneumonia patients. A retrospective case-control study on the risk factors associated with MDR *P. aeruginosa* mainly highlights the exposure to antibiotic classes as a major threat to MDR acquisition (Trinh et al. 2017). Carbapenem-resistant *P. aeruginosa* is currently listed as a “Priority 1: critical” pathogen by WHO

(Shrivastava et al. 2018). The antibiotic is a weak substrate to efflux pump and is less susceptible to hydrolysis by AmpC β -lactamases (Zhanel et al. 2014). Ceftolozane/tazobactam was also observed to retain its antibacterial activity against carbapenem and other antipseudomonal β -lactam isolates (Joshi et al. 2024). Despite ongoing trials evaluating various formulations, including those based on lipopolysaccharide, flagellum, alginate, whole-cell killed, and other OMP, there are still no licensed vaccines available against *P. aeruginosa*. The development of an effective vaccine has been challenging due to its ability to evade the host immune response. This complexity has hindered progress in creating protective vaccines, but researcher continues to explore innovative approaches to overcome this hurdle (Killough et al. 2022).

Acinetobacter baumannii

The *A. baumannii* is a Gram-negative coccobacillus and an opportunistic pathogen that has a high incidence among immunocompromised individuals, specifically those who are hospitalized for a prolonged period. It is most commonly seen in intensive care units globally and has rapidly become resistant to most antibiotics (Vijayakumar et al. 2024b). It is a major causative agent for fatal meningitis and pneumonia. *A. baumannii* can produce a number of virulence factors like capsular polysaccharides, phospholipids, lipopolysaccharides, toxins and protein secretion systems. The pathogenicity of the *A. baumannii* is enhanced by the Type II secretion system (T2SS) by involving LipAN (phospholipases C enzyme that degrades phosphatidylcholine), a major contributor to the colonization of the bacteria in the lungs (Li et al. 2023). The outer membrane protein A (OmpA) is the most abundant and known virulence factor of *A. baumannii* (Morris et al. 2019). Increase in OmpA expression is associated with several toxic effects, such as increased risk to pneumonia and bacteraemia (An et al. 2019; Geisinger et al. 2019; Uppalapati et al. 2020; Kyriakidis et al. 2021). *A. baumannii* translocated from the sputum and branch vascular regions causes infection and colonization in the lower respiratory tract. The bacteria adhere or colonize in the epithelial cells with the help of protruding structures such as pili, fimbrial CsuA/B, ABCDE-dependent pili (type IV secretion systems) and non-fimbrial autotransporter adhesions (type V secretion system) (Chen et al. 2022). The bacteria induce pro-inflammatory cytokines such as TNF- α , type I IFN and IL-1 β , which interfere with cell death and prolong pathogenicity (Zeng et al. 2020). Currently, recombinant vaccine strategies target single or multiple outer membrane proteins, which are easy to prepare and are safer as there is no risk of the pathogen reverting to its virulence and have far fewer adverse effects than other methods of

vaccination. Recombinant vaccines are designed to have a broad range of protection by selecting only those proteins that are conserved in strains of specific species of a particular genus (Singh et al. 2022).

***Mycobacterium* spp.**

Genus *Mycobacterium* consists of aerobic, nonmotile organisms that appear positive with acid-fast alcohol stains; it is a family of bacteria that can cause chronic lung infections and other respiratory infections; the two most widely known types are *Mycobacterium tuberculosis* (MTB), which causes tuberculosis (TB), and *Mycobacterium leprae*, which causes leprosy. TB is a highly contagious airborne disease and one of the top causes of death worldwide. MTB can stay dormant for years and persist in the body without any indication of illness, in which many people become asymptomatic carriers. It is transmitted through contaminated aerosol droplets, then travels through the respiratory tract and reaches the lungs, causing lung infections (Alsayed and Gunosewoyo 2023; R. and Anbarasu 2023).

Other species are classified as “nontuberculous” to set them apart, and lung infections can be caused by nontuberculous mycobacteria (NTM); the most common organism associated with NTM is the *Mycobacterium avium* complex (MAC), which has many subspecies including *avium*, *silvaticum*, *hominissuis* and *paratuberculosis* as well as *intercellular*, *arosiense*, *chimaera*. *Mycobacterium abscesses* are the third most common cause of lung infections (Johnson and Odell 2014). It is ubiquitous and can be mostly seen in water and soil sources; due to its biofilm formation, it is capable of resisting disinfectants and antibiotics. Hydrophobicity of NTM results in preferential aerosolization from water and along with this, it is resistant to high temperature and low PH. Infection of the lung is the most common clinical manifestation, which is termed pulmonary NTM disease (PNTM). Three forms of PNTM are described based on distinct pathology, which are fibro-cavitary disease, nodular bronchiectasis diseases, and hypersensitivity pneumonitis (Ratnatunga et al. 2020). Individuals who have immune suppression are prone to this infection. NTM unlike TB, person-to-person transmission has not been demonstrated. Mostly, the organism is ingested, inhaled, or implanted. The aerosolization of droplets small enough to enter the alveoli is considered to be the route of the acquisition of pulmonary disease (Matesanz López et al. 2021). Once the organism enters the individual, it usually settles in the lower respiratory tract. In some cases, the bacteria incite an inflammatory reaction with an influx in lymphocytes. The resulting release of cytokines and other mediators can lead to an infectious process that is seen as pneumonia. With an insufficient immune response to contain these *Mycobacteria*,

the pathogen can disseminate throughout the human host and cause extrapulmonary disease (Bethencourt Mirabal et al. 2024). *Bacillus Calmette-Guérin* (BCG) is a form of *Mycobacterium bovis* that prevents tuberculosis and other mycobacterium infections. It is the only vaccine against tuberculosis. It is also offered as a treatment against NTM infections. Researchers are currently conducting research on respiratory administration since natural infection and sensitization to *Mycobacterium tuberculosis* in humans tend to occur in the respiratory system (Okafor et al. 2024).

With the influence of multiple bacterial organisms involved in the pathogenicity of RTI, it has become quite

challenging for clinicians to prescribe appropriate antibiotics to be administered. Table 1 tabulates a detailed list of antibiotics along with their corresponding class and targets administered against RTI-associated bacteria.

Risk factors causing infections

Respiratory infections have multiple causes, including immunocompromised conditions, age-related specificity, and other comorbidities. Mostly, the weakened immune system a manifestation of conditions such on cancer, organ transplantation and HIV/AIDS are major risk factors for

Table 1 Summarized list of existing antibiotics and their targets associated to RTIs

Organism	Targets	Antibiotics	Classes	References
<i>Streptococcus pneumoniae</i>	PsaA and PspA	Vancomycin	Glycopeptides	(Jensen et al. 2015; El Ashkar et al. 2017; Cillóniz et al. 2018)
	<i>pbp1A, pbp1B, pbp2x, pbp2B, pbp2A, pbp3</i>	Penicillin	β -lactams	
	<i>ermB, (mefA, mefB, mefE)</i>	Penicillin, Cephalosporin	β -lactams	
	<i>tetM, tetO, tetK</i>	Erythromycin, Azithromycin	Macrolides	
<i>Haemophilus influenzae</i>		Tetracycline, Doxycycline, Minocycline	Tetracycline	(Kaczmarek et al. 2004; Tristram et al. 2007; Abdou 2023)
	<i>gyrA, parC</i>	Ciprofloxacin	Fluroquinolones	
	<i>blaTEM, blaROB-1, pbp4, pbp5</i>	Penicillin, Cephalosporin, Carbapenem	β -lactams	
	<i>ermB, mefA</i>	Macrolides, Erythromycin, Azithromycin	Macrolides	
<i>Staphylococcus aureus</i>	<i>cat</i>	Chloramphenicol	Amphenicol	(Dörries et al. 2014; Nandhini et al. 2022)
	<i>pbp3</i>	Penicillin	β -lactams	
	<i>pbp2a, mecA, blaZ</i>	Tetracycline	Tetracycline	
	<i>tetK, tetL</i>	Trimethoprim	Sulphonamides	
<i>Streptococcus pyogenes</i>	<i>dfrA, dfrK, ermC, ermA</i>	Clindamycin, Erythromycin	Macrolides	(Berbel et al. 2021)
	<i>prtF1</i>	Erythromycin	Macrolides	
	<i>pbp2x, pbp1A</i>	β -lactams	β -lactams	
	<i>erm(A), mef(A)</i>	Macrolides	Macrolides	
<i>Klebsiella pneumoniae</i>	<i>tet(O)</i>	Tetracycline	Tetracycline	(Pulzova et al. 2017; Basu et al. 2024; Huy 2024)
	<i>blaSHV, blaTEM, blaCTX, blaLUC5, blaSFO, blaGES,PER, blaVEB, blaTLA, blaKPC, blaNDM, blaVIM, blaIMP, blaOXA, blaCMY, blaDHA, blaFOX, blaMOX</i>	β -lactams	β -lactams	
	<i>AcrAB-ToC, oqxa, RARA, ramA, ramR, AcrR, rpsJ, 16 S rNA methylase, tetA</i>	Tigecycline	Tetracycline	
	<i>DNA gyrase, topoisomerase IV, mpK36, acrAB, kdeA, OqxAB, and aa(6')-Ib-cr</i>	Quinolone	Quinolone	
<i>Pseudomonas aeruginosa</i>	<i>PhoPQ, pmrA, pmrD, mcr-1, and mgrB</i>	Polymyxin	Polymyxin	(Martin et al. 2018; Hibbert et al. 2023)
	<i>fos</i>	Fosfomycin	β -lactams	
	<i>AmpC-Betalactamase, oprD</i>	Aztreonam, Ceftazidime, Meropenem, Timentin	β -lactams	
		Clarithromycin, Azithromycin	Macrolides	
<i>Mycobacterium</i>	<i>erm</i>	Tetracycline	Tetracycline	(Luthra et al. 2018; Miryala et al. 2021)
	<i>tet0, tetX</i>			
<i>Acinetobacter baumannii</i>	<i>MexXy-OprM</i>	Tobramycin/ Ciprofloxacin	Aminoglycosides/ Quinolone	(Černiauskienė et al. 2023; Shi et al. 2024)
	<i>gyrA, parC</i>	Quinolone	Quinolone	
	<i>pmrA</i>	Colistin	Polymyxin	
	<i>CarO</i>	Carbapenem	β -lactams	
	<i>TEM, SHV, CTX-M, PER, VEB, KPC, GES-22, CarO</i>	Penicillin, Cephalosporins, Carbapenems	β -lactams	

pneumonia as well as other respiratory infections (Azoulay et al. 2020). Age groups under 5 year and elderly over 65 years are at risk due to age-related changes in lung structure and immune function (Connor et al. 2022). Underlying medical conditions like diabetes, chronic lung diseases and cardiovascular diseases can increase the risk and severity of infections, particularly in elderly populations (Sethi 2010).

Immunocompromised individuals

The major risk of morbidity and mortality associated with infectious diseases is dependent upon the individual immune response. In immunocompromised individuals, the rate of these infections is higher due to poor immune responses. The continuing decline of immune responses is often associated with severe ailments such as cancer, AIDS and their related treatments. Often in such patients, due to the non-specific treatment strategies available for such diseases, the immune system of these patients is severely compromised, leading to making these patients more vulnerable to infectious diseases. These infections that occur in patients are mostly due to immune deficiency, and they may vary according to the patient. (Algar and Novelli 2007). Pulmonary infections are quite common in immunocompromised patients due to their constant environmental exposure. Some common RTI-associated pathogens that affect immunocompromised individuals include *S. pneumoniae* and *H. influenzae* and non-fermenting Gram-negative bacteria which cause pulmonary infections along with the humoral immunity deficiency. As reported by Aleem and team, the respiratory tract is continuously exposed to unsanitary environments that have adverse effects on human health, especially in the case of COPD and asthma (Aleem et al. 2024). Corticosteroids and low CD4 expression increase the rate at which various pathogens colonize the respiratory tract, also changing the microbiota (Letourneau et al. 2014). The colonization and change in the microbiome, in turn, results in dysbiosis and impaired immune monitoring. The large intake of inoculum with pathogenic agents in the alveolar space or the immunodeficient system consisting of pro-inflammatory cytokines dominantly majorly triggers pneumonia (Sattar and Sharma 2024).

Age factor

Age groups are a significant risk factor associated to RTIs, as the increase in the risk of infections is mostly seen in infants and elder age groups. These can be due to the fact that these age groups are more vulnerable to malnutrition, poor immunization, existing chronic conditions and natural pollution (air pollutants) (Abayneh et al. 2022). As reported by Childs and his research group, the critical burden for RTIs

is mostly predicted to be twice for the age group > 65 years in 2050 when compared to that of in 2012. Older adults in long-term care centers are also prone to the risk of acquiring respiratory infections. The combination of closeness among the residents and the advancement of their age is a major cause for infections and complications (Childs et al. 2019).

Comorbidities

An important risk associated with RTIs was witnessed during the COVID-19 pandemic spanning from 2020 to 2021. COVID-19, which is caused by severe acute respiratory syndrome coronavirus-2 (SARS CoV-2) virus belonging to the family of Coronavirus, order Nidovirales, (Lotfi et al. 2020). Viral infections frequently impact the upper and lower respiratory tracts. Influenza A, B viruses, and respiratory syncytial virus (RSV) are primary contributors to mortality among hospitalized individuals. Emerging respiratory viruses with pandemic potential, such as the Middle East Respiratory Syndrome (MERS) and Severe Acute Respiratory Syndrome (SARS) coronaviruses, have zoonotic origins (Peltola et al. 2020). These viruses predominantly affect individuals aged < 5 years and > 65 years, as well as pregnant women and immunocompromised individuals who are particularly vulnerable. The transmission of these viruses primarily occurs through three routes: direct or indirect contact with contaminated surfaces or objects (fomites), short-range droplet-borne transmission, and aerosol transmission (He et al. 2023). The symptoms vary according to the host; for instance, in human adults, the symptoms include fever, cough and shortness of breath, while, for the elderly age group, it is mostly caused by immunosuppression, which can lead to fatal conditions (Sharma et al. 2021). Comorbid pulmonary conditions are also another major challenge, leading to some severe conditions such as asthma or COPD diseases that can, in turn, lead to RTIs. Under normal conditions, the tracheal or parenchymal cells help prevent any microbial eruptions, however, individuals with COPD have been observed to be more vulnerable to airway mucosal infections caused by pathogenic bacterial and viral agents. Such infections proceed with a high risk of inflammation in patients, which can cause increase severity (Lange 2009).

Existing antibiotic resistance mechanisms

The inappropriate use of antibiotics leading to antimicrobial resistance is a continuous global threat. Various mechanism permits the pathogenic bacteria to resist conventional antibiotics, such as the efflux pumps, increasing the permeability of the cell membrane, drug target modifications and inactivation of antibiotics by producing key molecules (Miriti et al. 2023). With the misuse of these antibiotics when acquired

without a proper prescription, the threat of pan resistance is developed. The antibiotics which are prescribed inappropriately are also a leading cause of resistance.

After the first antibiotic, penicillin, was discovered, various other antibiotics, including natural and synthetic antibiotics, were used to reduce the bacterial burden and lower the growth of bacteria. In addition, many antibiotic resistance mechanisms in the bacterial metabolism process have additional functions. The efflux pump has the ability to evacuate toxins, like heavy metal ions, from the cell membrane while simultaneously transporting particular antibiotics (Du et al. 2018). Antibiotics cause the bacteria to evolve related mechanisms to withstand their action continuously. The bacterial defense system against antibiotics is made up of these new resistance mechanisms that researchers have found. These mechanisms include going into dormancy, secreting proteins that protect targets, controlling metabolism, and starting self-repair processes (Fig. 2). These mechanisms collectively make up the bacterial defense system against antibiotics (Zhang and Cheng 2022).

Modification of drug target

Bacterial cells have different components that may act as a target for different anti-bacterial agents. These targets are modified at regular intervals in order to enable resistance mechanisms to the drug. The key resistance mechanism

against β -lactam antibiotics is altering the structure of their drug target, that is, penicillin-binding proteins (PBPs). The PBPs are transpeptidase proteins that are involved in the construction of cell walls. Any alterations in the PBPs can directly affect the binding of the antibiotics. Additionally, these structural changes will also aid the bacteria in overcoming the antibacterial effects due to the poor binding affinity of the antibiotics (Reygaert 2018). Similarly, the bacteria also promote drug resistance by altering metabolic pathway proteins, like that observed in *S. pneumoniae*, *H. influenzae*, and *P. aeruginosa* (Guitor and Wright 2018), where mutations in dihydropteroate synthase (DHPS) and dihydrofolate reductase (DHFR) involved in folate synthesis aiding in drug resistance against trimethoprim (Hawser et al. 2006).

Limiting the uptake of the drug

Bacteria have a natural tendency to limit drug absorption, where the composition of the cell wall plays an important role. The peptidoglycan layer of Gram-positive (thick-layered) and Gram-negative bacteria (thin-layered) allows only certain types of molecules to pass through the cell membrane (Reygaert 2018).

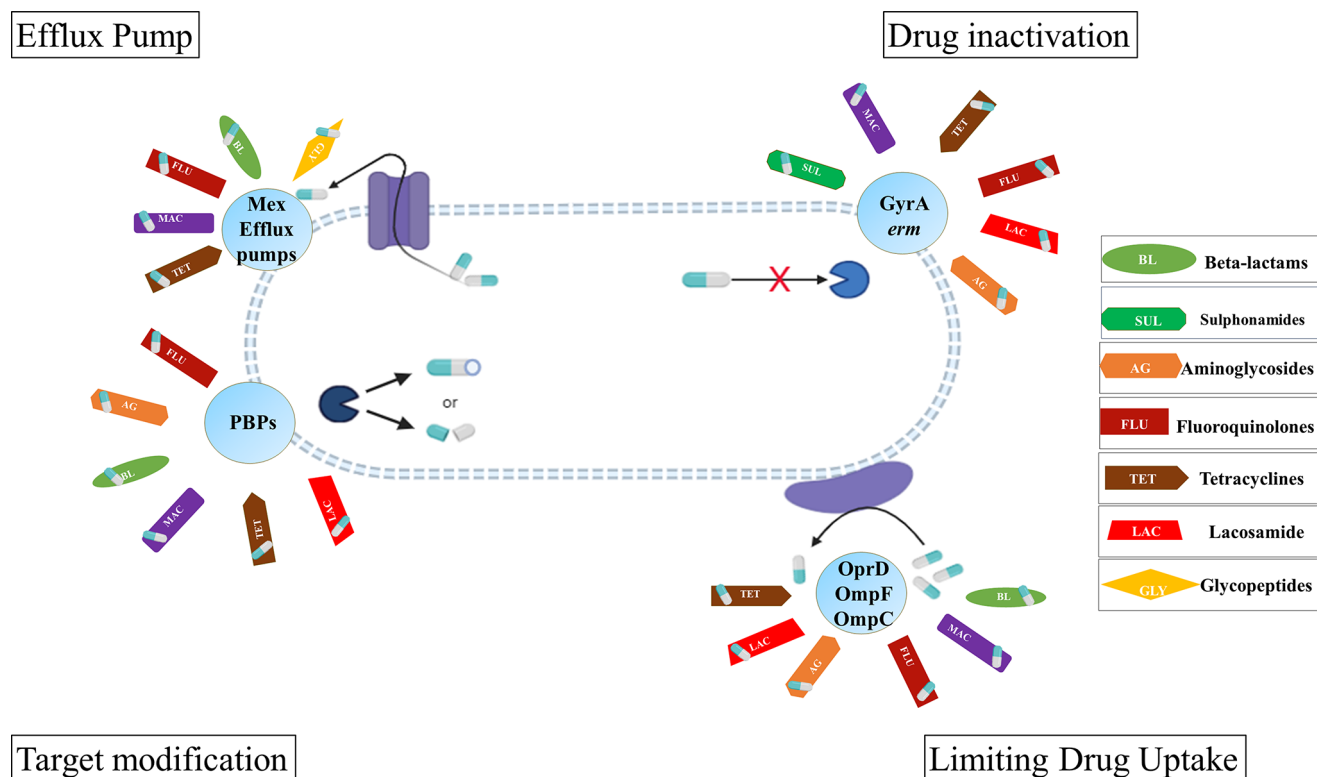


Fig. 2 Overall resistance mechanisms and the antibiotics related to the cause of RTIs

Efflux pump

The efflux of antimicrobial agents is an alternative major mechanism aiding in drug resistance. The membrane proteins such as (OprM, OprJ, and OprN) while maintaining the intracellular concentration of the antibiotics, also export the antibiotics from the cell before they reach their targets. The efflux pumps present in the cytoplasmic membrane are susceptible to all antibiotics except polymyxin (Kapoor et al. 2017). Currently, there are five main families of efflux pumps in bacteria, and they are classified on the basis of structure and energy. ATP binding cassette (ABC) proteins use energy derived from ATP hydrolysis, while the multidrug and toxic compound extrusion (MATE) proteins use Na^+ gradient as an energy source and efflux cationic dyes. Small multidrug resistance (SMR) families are energized by proton-motive force H^+ , and these are hydrophobic in nature. The major facilitator superfamily (MFS) catalyses the transport via solute/cations (H^+ or Na^+) symport or solute/ H^+ antiport. The resistance-nodulation-cell division (RND) family catalyses substrate efflux via a substrate/ H^+ antiport mechanism that is predominant in gram-negative bacteria (Poole 2007; Reygaert 2018).

Inactivation of drug

There are two main ways to inactivate the drug inside the bacteria. It can be through the degradation of the antimicrobials or by chemically modifying the drug molecule. This can occur by the transfer of a chemical group to the drug, where it interacts with the drug target and renders the drug inactive. The common chemical modifications include

phosphorylation, acetylation, hydroxylation, adenylation and glycosylation. The β -lactam antibiotics (such as penicillin, cephalosporins, carbapenems, and monobactams) are often subjected to the former mechanism, where potent β -lactamases are produced by the bacteria, which cleaves the β -lactam ring in the antibiotics. The common resistance mechanism observed against macrolides is the chemical modification using esterase and phosphotransferase enzymes. While tetracycline antibiotics are subjected to hydrolysis using the tetX gene in *P. aeruginosa*, aminoglycosides are often inactivated by phosphorylation and adenylation (Reygaert 2018; Ali et al. 2023).

Table 2 Provides an overview of the antibiotics associated with RTIs and the increasing prevalence of drug resistance among the RTI pathogens.

Therapeutic solutions

The most prevalent therapeutic solution for circumventing the pathogens that were acquired from community spread or hospital-borne is the prophylactic application of antibiotics. However, with the increase in antibiotic resistance globally, it has become imperative to identify alternative therapeutic regimens. These regimens need to overcome the existing challenge of drug resistance and also provide good clinical translation values. The various suggested alternatives include the utilization of phage-driven therapeutic proteins for phage therapy, the identification of novel drug targets and drug repurposing (Manohar et al. 2020).

Table 2 List of antibiotics showing resistance and susceptibility against the RTI pathogens

Organisms	Antibiotic Resistance	Antibiotic Susceptible	Reference
<i>Streptococcus pneumoniae</i>	Macrolides, Gentamicin, Isoniazid, Colistin, Sulphonamides, Tetracycline, Cephalosporin	Penicillin, Fluoroquinolones, Ceftriaxone, Vancomycin, Levofloxacin, Meropenem, Linezolid	(Varghese et al. 2022; Basu et al. 2022; Amari et al. 2023; Sandoval et al. 2024)
<i>Haemophilus influenzae</i>	Ampicillin, Erythromycin, Cefuroxime, Ciprofloxacin, Cefaclor	Cefotaxime, Clarithromycin, Azithromycin, Fluoroquinolones, Tetracycline,	(Ekinici et al. 2023; Abavisani et al. 2024)
<i>Staphylococcus aureus</i>	Amoxicillin, Oxacillin, Erythromycin, Chloramphenicol	Amikacin, Tetracycline, Cefoxitin, Levofloxacin, Imipenem, Vancomycin	(Ragunathan et al. 2019; An et al. 2024)
<i>Pseudomonas aeruginosa</i>	Ciprofloxacin, Aztreonam, Aminoglycosides, Fluoroquinolones, Carbapenems, Tetracyclines	Doxycycline, Ceftazidime, Ceftriaxone, Colistin	(Sastre-Femenia et al. 2023)
<i>Klebsiella pneumoniae</i>	Ampicillin, Amoxicillin, Ceftazidime, Cefotaxime, Cefepime, Tetracycline	Amikacin, Meropenem, Cefepime, Ertapenem, Levofloxacin, Ceftazidime, Ciprofloxacin, Cefuroxime	(Vijayakumar et al. 2024a; Vollset et al. 2024)
<i>Streptococcus pyogenes</i>	Streptomycin, Ceftriaxone, Vancomycin,	Penicillin, Cefotaxime, Chloramphenicol, Levofloxacin	(Kebede et al. 2021)
<i>Acinetobacter baumannii</i>	Penicillin, Aminoglycosides, Fluoroquinolones, Carbapenems, Gentamicin, Polymyxins	Amikacin, Colistin, Tigecycline, Ciprofloxacin, Ceftazidime, Cefepime	(Shi et al. 2024)
<i>Mycobacterium</i>	Rifampicin, Isoniazid	Azithromycin, Amikacin, Ciprofloxacin, Clarithromycin, Ethambutol, Streptomycin, Isoniazid	(Luthra et al. 2018)

Targeting immune receptors

The natural innate system accounts for the most reliable defense mechanism against pathogenic organisms. The innate immune cells at the molecular level detect the presence of microorganisms using the pattern recognition receptors (PRRs) that detect the evolutionary conserved structures on pathogens known as pathogen-associated molecular patterns (PAMPs) (Behzadi et al. 2021). The lipids, proteins, or nucleic acid, along with the host signature molecules, aid the receptors in activating the defense mechanisms by producing the pro-inflammatory cytokines, which alert the host cells of the presence of infection. The respiratory tract's innate immunity relies on a multi-layered defense system, starting with a mechanical barrier that prevents bacterial invasion and exposure to foreign microorganisms. The barrier consists of three main components: the first is an upper layer of mucus that physically blocks pathogens. Second, a lower layer of mucus contains various antimicrobial substances, such as cytokines and TLR molecules, which are secreted by epithelial cells to combat infections. Third, efficient mucociliary clearance works in conjunction with immunoregulatory factors to remove pathogens and coordinate immune responses (Grzela et al. 2012). Cytokines play a key role in this process, functioning through endocrine, autocrine and paracrine mechanisms. In addition, mononuclear cells in human peripheral blood express a wide range of immunoproteins and immunomodulators, including adhesins, transcriptomic molecules, soluble receptors, cytokine growth factors, chemokines and activated endothelial molecules, particularly in patients with autoimmune diseases (Behzadi et al. 2022b). The adaptive immunity involves the defense mechanism based on the foreign particles (antigens). The cells carry the receptors that identify the foreign molecule pattern but do not identify the pathogens. The approaches that are currently being investigated target the innate immune PRRs such as Toll-like receptors (TLRs) and NOD-like receptors (NLRs). The significance of targeting TLR signalling for bacterial defense is supported by the fact that modulation in TLR signalling has been reported to cause impaired control of pathogenic infections. There are mainly two major ways in which the TLR signalling is altered, namely by the production of agonists and antagonists' molecules. In vitro studies have shown that the stimulation of TLR2 and TLR4 increases the various cytokines production by the epithelial cells. The former works by producing an agonist that often mimics the natural ligand, thereby enhancing inflammation by activating the signalling pathway. The latter mechanism involving an antagonist deal with the production of molecules that can significantly suppress the immune pathways and associated harmful inflammations. However, the latter mechanism can potentially

repress the protective mechanism of the infected host. NLRs belong to the cytoplasmic innate immune receptors that are able to detect the microbial as well as the dangerous endogenous molecules that can induce inflammation and also direct antimicrobial activities (Hancock et al. 2012).

Phytocompounds

Plants are well known for their therapeutic potential in various forms, such as raw extract or powders, or also in purified products. The various antimicrobial properties possessed by these compounds opens avenues of research for their use against the treatment of infections caused by MDR organisms (Shriram et al. 2018; Anand et al. 2019). Currently, very few phytochemicals such as codeine, capsaicin, reserpine, colchicine have been approved by the Food and Drug Administration (FDA) for their therapeutic application (Kongkham et al. 2020). Latest report suggests the use of phytochemicals to escalate the efficiency of existing antibiotics. For instance, the flavonoid kaempferol and its natural glycoside derivative from perse lineage have been shown to enhance the antimicrobial activity of ciprofloxacin in NorA overexpressed *S. aureus* strains (Holler et al. 2012). Essential Oils (EOs) have also been extensively studied for their remarkable antibacterial properties against a wide range of pathogens, including Gram-positive and Gram-negative bacteria. It is noteworthy that essential oils exhibit a dual mechanism of action, namely to kill bacteria and also to enhance the effectiveness of antibiotics against pathogenic microbes. Nevertheless, the susceptibility of EOs to degradation hinders the stability of the molecule, which plays a crucial role in determining the translational benefits of the compounds. EOs are considered to be safe for consumption of vital host tissues; therefore, any chemical modification increasing the stability of the compounds can hugely benefit the translation effects (Khare et al. 2021).

Phage therapy

Phage therapy is a promising strategy used against respiratory infections, particularly those caused by antibiotic-resistant strains. Inhaled phage therapy has shown significant potential in the prevention and treatment of these infections (Chang et al. 2018). Phage works by infecting and replicating with the targeted bacteria, which leads to a rapid decrease in bacterial populations (Liu et al. 2016; Chang et al. 2018). Clinical trials have demonstrated its safety and tolerability, making it a promising alternative treatment option. Nebulization is the primary method for phage aerosolization, where liquid phage formulations are converted into fine droplets using commercial nebulizers (Golshahi et al. 2008; Chang et al. 2018). Recent studies have demonstrated the

efficacy of phage therapy in combating antibiotic-resistant bacteria. Teng and team conducted an in vitro study using a mouse model, showing that phage therapy reduced drug resistance in *S. aureus* compared to traditional antibiotics (Teng et al. 2022). Similarly, Prazak found that phage therapy significantly reduced mortality in a lethal model of *K. pneumoniae* and *A. Baumannii* compared to a placebo (Prazak et al. 2019). Phage lysin, such as Lysin Cpl-1, has also shown promising results in treating lung infections. These phage-encoded proteins can hydrolyse peptidoglycan in the bacterial cell wall, displaying high antimicrobial effects without inducing bacterial resistance. Several endolysins have been characterized and tested against *S. pneumoniae* in both in vitro and in vivo studies. Notably, Lysin Cpl-1 has demonstrated therapeutic potential in various animal models of infection when combined with antibiotics or used in conjunction with other lysins (Chang et al. 2018).

Preclinical studies in animal models have shown the efficiency of phage therapy in treating respiratory infections. High doses of phage administered shortly after bacterial challenge have provided optimal protection against acute infections caused by *P. aeruginosa* (Fothergill et al. 2014; Chang et al. 2018). Challenges such as formulation stability during neutralization and storage, as well as the development in the resistance, remain an area of concern. Strategies like phage therapy training may help address the resistance issues and prolong the effectiveness of phage therapy (Pabary et al. 2016; Chang et al. 2018).

Active and passive immunization

Immunization has played an important role in the control and spread of infectious diseases, thereby preventing outbreaks and epidemics. Both active and passive immunization were helpful in controlling the spread of infections, particularly viral and bacterial infections. The vaccine subunit for *A. baumannii* has been explored and the results indicate protective benefits of passive immunization by yolk antibodies (IgY) in mice model studies; mice actively immunized with outer membrane protein A (OmpA) had 40% survival rates, but interestingly in passively immunized mice the survival rates increased to 90%. However, active immunization with recombinant *A. baumannii* (Trx-outer membrane protein W) Trx-OmpW protein resulted in 100% protection in mice, while passive immunization was 83%. On the whole administering both passive and active immunization with multiple vaccines and antibodies combination procedures results in better survival rates (Yang et al. 2023). Inoue et al., reported protective immunization effects of *P. aeruginosa* V antigen (PcrV) targeting the type III secretion systems (T3SS) along with passive immunizations blocked the PcrV antigens (Inoue et al. 2023). In another study

conducted against the T3SS of *P. aeruginosa* using derived recombinant PcrV protein; the immunized mice survived lethal challenge infections with *P. aeruginosa*. Together with active immunization a positive correlation between the survival rate of the mice and the polyclonal PcrV antibody was observed in various animal models for chronic bacterial pneumonia (Naito et al. 2017). Detailed Studies related to RTI pathogens are mentioned in Table 3.

Computational approaches in comprehending RTIs

High throughput computational advancements like whole genome sequencing, parallel sequencing and single-cell sequencing have generated enormous data that, on processing, identifies key alterations observed in microorganisms. Identifying these changes is critical in understanding the evolution of the organism and its pathogenicity. The computational approaches to classify, characterize and design novel therapeutics for RTIs can be broadly classified into three approaches, namely genomics, systems biology and structural bioinformatics studies. A detailed overview of the impact of these technologies has been discussed below:

Genomics advancements in RTIs

The identification and characterization of respiratory pathogens have been completely transformed by whole genome sequencing (WGS), eliminating the need for traditional culture-based techniques and enabling precise identification of bacterial, viral and fungal pathogens. Antibiotic resistance genes, certain virulence factors, and other genetic variables that contribute to the pathogenicity of these organisms can be found through genomic analysis (Ellington et al. 2017). With AMR in respiratory pathogens becoming a significant public health problem, identifying known resistance genes and novel mutations to various antibiotics, WGS has emerged as a powerful tool for detecting and tracking antibiotic-resistant mutations in these pathogens (Deng et al. 2016). These data are crucial for developing appropriate antibiotic treatment regimens and implementing efficient infection control strategies.

Outbreak and surveillance investigations of respiratory infections now rely heavily on genomic epidemiology using WGS data. Researchers can trace the origin of an outbreak, identify the transmission patterns, and track the pathogens evolution over time by comparing the genome sequences of isolates from different cases or geographic areas. This information can be used to implement target control measures and prevent further spread (Grad and Lipsitch 2014). The molecular mechanism behind the virulence and pathogenesis of respiratory pathogens have been widely understood by comparative genomics and functional genomics

Table 3 List of different modes of immunization (active and Passive) against the RTIs pathogens

Organism	Active	Passive	Reference
<i>Streptococcus pneumoniae</i>	Mice immunized with purified LytA similar to pneumolysin	Monoclonal and polyclonal antibodies to PspA protect mice from fatal bacteraemia	(Briles et al. 2019)
<i>Haemophilus influenza</i>	cCHIP-P6 nanogel nasal vaccine induced P6 specific mucosal IgA and serum IgG responses in the absence of active adjuvant	Immune serum transferred to juvenile chinchillas by intracardiac injection by serum pools: human anti-tetra-Pn, human anti-MeAC-Hib and normal adult serum	(Novotny et al. 2006; Naka-hashi-Ouchida et al. 2022)
<i>Staphylococcus aureus</i>	SA4Ag targeting conjugated Cp5/CP8 plus recombinant ClfA/MntC StaphVax targeting Iron surface determinant B (IsdB)	SEB-specific antibody using chicken immunoglobulin IgY targeting Staphylococcal enterotoxin B Tosatoxumab or AR-301 using fully human monoclonal IgG1 antibody targeting alpha toxin	(LeClaire et al. 2002; Hassan-zadeh et al. 2023; Scafa-Udriste et al. 2023)
<i>Klebsiella pneumoniae</i>	Five <i>K. pneumoniae</i> outer membrane proteins formulated with aluminum hydroxide gel adjuvant into BALB mice	Anti OmpA- CPS19 antibodies partially protect against pneumococcal infections	(Zhang et al. 2021; Assoni et al. 2021; Dou-radinha 2024)
<i>Streptococcus pyogenes</i>	Mice immunization with three intramuscular doses of VAX-A1 in alum was shown to provide complete protection The heterologous serotype of group A <i>Streptococci</i> with a vaccine consisting of widely shared C-repeat region of the M6 protein	Live <i>Streptococci</i> mixed affinity purified M-protein specific sIgA or IgG antibodies and administered to the animals The protection against <i>Streptococcal</i> pharyngeal colonization was also shown by the oral administration of purified lipoteichoic acid (LTA) but not the deacylated LTA before oral streptococcal challenge in mice	(Fischetti 2019; Gao et al. 2021)
<i>Pseudomonas aeruginosa</i>	Aerugen (Octavalent OPS-Toxin A conjugate) Pseudogen (Heptavalent O- antigen)	PsAer-IgY (anti- <i>Pseudomonas</i> IgY gargle) KB001-A (anti-PcrV PEGylated mouse Mab)	(Zuercher et al. 2006; Lazar et al. 2009; Jain et al. 2018)
<i>Acinetobacter baumannii</i>	OmpA is found to be a predominant target of humoral immunity during sublethal infection in mice. Mice immunized with BamA produced IgG with high titers. 80% survival rate	Mice immunized with IgY targeting Bap subunit Intraperitoneal injection of serum Blp-1 immunized mice provided complete protection to mice	(Singh et al. 2017; Ranjbar et al. 2022; Yang et al. 2023)
<i>Mycobacterium</i> spp.	ID-93 + GLA-SE was used as the therapeutic vaccination in mouse models and in cynomolgus macaques	Sera obtained from mice treated with RUTI, protection against reactivation of <i>M. tuberculosis</i> in SCID mice	(Guirado et al. 2006; Bouzeyen and Javid 2022)

approaches (Patarčić et al. 2015). Genes and signalling pathways involved in processes such as adhesion, invasion, toxin production and host immune evasion can be found by comparing and analysing the genomes of virulent and non-virulent strains. The development of innovative therapeutic approaches and prevention measures can be influenced by this knowledge.

The role of host genetic variables influencing disease severity or susceptibility to respiratory infections has also been investigated through genomic studies. Genetic variations associated with an increased risk of serious respiratory infections or complications have been identified through genome-wide association studies and other genomic techniques (Costain et al. 2021). Metagenomics is another method used to study a specific community of microorganisms. For RTIs, respiratory metagenomics (RMg) is a diagnostic test that quickly identifies the potential pathogens and antimicrobial resistance determinants. Chronic infections caused by *P. aeruginosa* in cystic fibrosis patients have shown microbial dysbiosis that shares core gene

profiles and antimicrobial resistance genes (Edgeworth 2023). Metagenomics next-generation sequencing (mNGS) is a new diagnostic method that has been employed to identify rare pathogens and emerging agents. This method can be applied to different specimen types, including sputum, bronchoalveolar lavage fluid (BALF) and pleural effusion. It detects a wider range of potential pathogens with higher accuracy and in a shorter duration of time (Liang et al. 2022). The metagenomics approach includes two methods; the first includes deep amplicon sequencing that focuses on the specific taxonomic marker, such as the 16 S rRNA gene. The second method is the shotgun metagenomics method, where the random fragments of all genomes are taken and sequenced. The fragments are assembled to reconstruct the genomes in the sample. In shotgun metagenomics, the sequencing can be done immediately after the enrichment step, such as a capture-based method to increase the detection of the relevant organisms (Miller et al. 2013).

Systems biology advancements in RTIs

Systems biology approaches are increasingly being used to investigate respiratory infections. These approaches enable a comprehensive understanding of the complex molecular networks and host-pathogen interactions that play a role in these infections. Systems biology studies integrate multiple omics data and computational modelling. Researchers have studied the global gene expression profiles of respiratory pathogens during infection using transcriptomics and genomics methods (Li et al. 2022). For instance, virulence-associated genes and regulatory networks in *P. aeruginosa* and *K. pneumoniae* have been found through RNA-seq studies, which has contributed to the discovery of possible therapeutic targets (Chakravarty and Massé 2019). Proteomics and metabolomics techniques were used to study protein expression and metabolic changes in respiratory pathogens at different stages of infection. These studies have shed light on the metabolic pathways and adaptive mechanisms necessary for the virulence and survival of the bacteria (Mücke et al. 2020).

In addition, host-pathogen interactions and the host immune response in the RTIs have been studied using systems biology pipelines. A comprehensive study of transcriptomic and proteomic information derived from infected host cells and tissues has revealed the complex host defense mechanisms, as well as tactics pathogens, use to evade these responses (Dhillon et al. 2020). Furthermore, various omics data have been integrated, and a system-level understanding of bacterial respiratory infections achieved through the use of network analysis (Bevivino et al. 2019). These methods have been applied to create metabolic models, host-pathogen interaction networks, gene regulatory networks, and signal transduction networks, which help in discovering possible therapeutic targets and creating more effective treatment plans (Anitha et al. 2016; Panditrao et al. 2022).

Structural bioinformatics advancements in RTIs

Numerous structural biology techniques have been used to study respiratory tract infections providing important new insights into the molecular mechanisms of pathogenesis, antibiotic resistance, and possible treatments. To understand the molecular basis of resistance and guide the development of new antibiotics or strategies to overcome resistance, X-ray crystallography has been used to determine the structures of antibiotic-resistant enzymes and cell wall alterations (Rostock et al. 2018; Conn et al. 2019). Elucidating the structures of bacterial virulence factors such as adhesins of *S. pneumoniae* technique (Jedrzejak 2001). The dynamics and interactions of bacterial proteins involved in respiratory infections have been successfully studied using

NMR spectroscopy. NMR spectroscopy has also been used to study the interactions between bacterial virulence factors and host proteins, contributing to the understanding of host-pathogen interactions and the development of possible therapeutic interventions (Biron et al. 2011). The technique has been used to study the structural dynamics of bacterial toxins such as *P. aeruginosa* exotoxin A and to provide insight into their mechanisms of action and possible inhibition strategies (Popoff 2024).

An effective technique to study the structures of large macromolecular complexes involved in respiratory infections is Cryo-Electron Microscopy (Cryo-EM). This technique was used to study the structures of bacterial secretion systems, which are crucial for the pathogenicity of respiratory pathogens such as *P. aeruginosa* (Arif et al. 2022). In addition, bacterial pili and other surface appendages involved in host cell adhesion and invasion during respiratory infections have been studied using Cryo-EM (Hilleringmann et al. 2008). Karlsson and his team's study stated the application of nano-liquid chromatography (LC) coupled with tandem Mass spectrometry (MS) and a bioinformatics pipeline, utilizing high-quality curated reference genome sequences to identify species unique peptides as potential biomarkers for bacterial pathogens in respiratory infections. These peptide biomarkers can be employed in the development of a targeted MS approach offering higher specificity and sensitivity compared to discovery-based studies, with advantages in MS analysis and data evaluation (Karlsson et al. 2020). To ensure the proper quantitative analysis of *de novo* protein-protein interaction (PPI), emerging computational methods like machine learning (ML) have been used alongside structural feature-based PPI (protein-protein interaction) prediction techniques. It helps in the prediction of microbe-host molecular interactions. Some of the *de novo* PPI predictions include protein properties such as post-translation modifications, chemical composition, tissue distribution, molecular weight, homology to human binding partners, and relevance of protein in the host network. Studies employing individual ML models such as 12 regularised logistic regression, random forest (RF), support vector machine (SVM), Artificial Neural Networks (ANN) and K-Nearest Neighbours (K-NN) have accurately predicted bacterium-phage interactions (Sudhakar et al. 2021).

Drug discovery is another major application of the structural bioinformatics approach, where filtered pharmacokinetic-pharmacodynamic potent small molecules or repurposed drug molecules can be used to study the inhibitory effects on target proteins. This can be achieved using molecular docking studies and the stability and energy trajectories can be studied using molecular dynamics simulations. Figure 3 depicts an overall schematic representation compiling the genomics, systems biology and structural

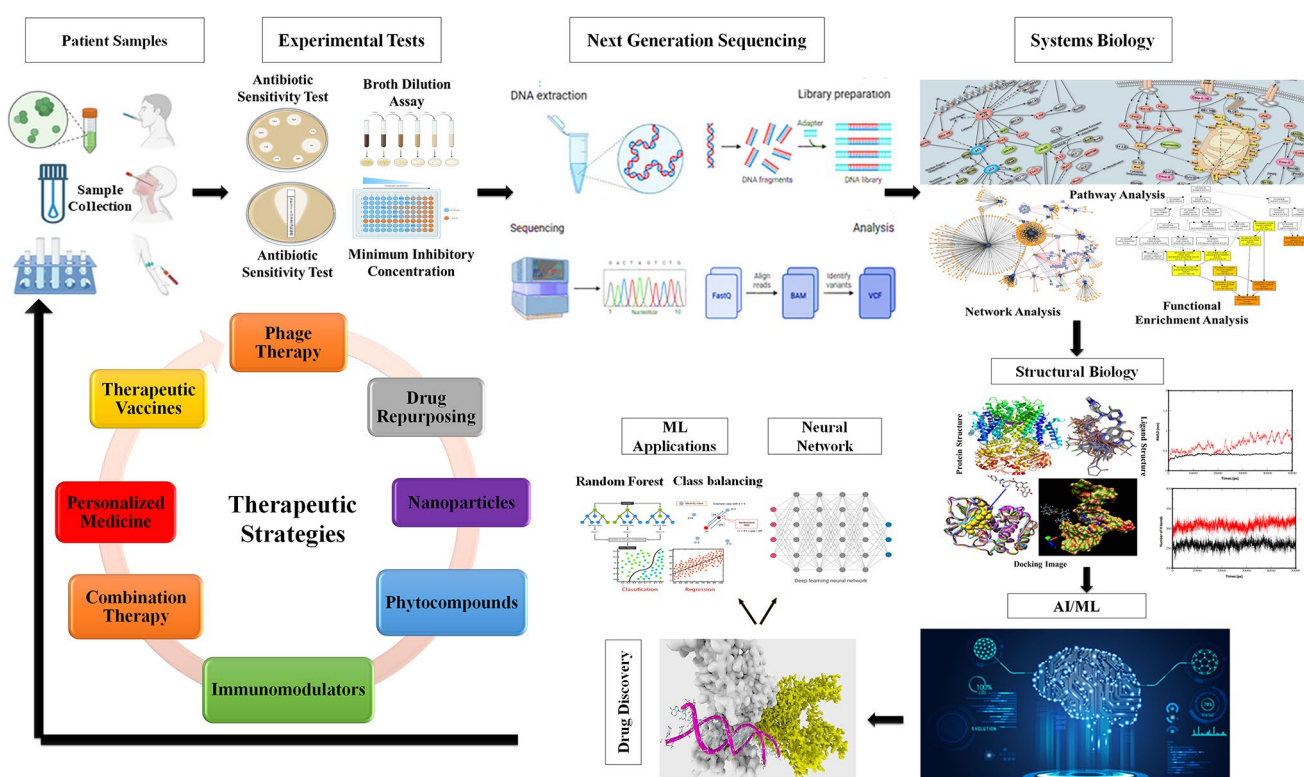


Fig. 3 Alternative strategies and future prospects for understanding rtis and their treatment

bioinformatics approaches compiled with artificial intelligence to manage and control RTIs.

Immunoinformatics in RTIs

A vast array of immunoinformatics studies has helped in the better understanding of disease pathogenesis, diagnosing the immune system dynamicity during bacterial infections. Immunoinformatics techniques are used to reduce the time and expense of vaccine design. Bioinformatics approaches involving reverse vaccinology strategies are being widely used for the development of multi-epitope vaccines (Roy et al. 2024; Biswas et al. 2024). The multi-epitope vaccines are designed with antigenicity, non-toxicity, and allergenic properties. The vaccine constructed can be utilized to evaluate the simulated immunogenic response of the host (Jafari et al. 2020). Multi-epitope vaccines (MEVs) are used in immunoinformatic methods, such as predicting epitopes with reliable servers and attaching epitopes with linkers and adjuvants (Priyamvada et al. 2024). MEVs have shown potential against invasive *H. influenza* infections (AlChalabi et al. 2022). In the case of *S. pneumoniae*, the best antigenic protein *pspA* and epitopes were identified to design a multi-epitope vaccine comprising cytotoxic T lymphocytes (CTL), Helper T lymphocytes (HTL), and linear B lymphocytes (LBL) epitopes. These epitopes are

joined using linkers such as KK, CPGPC, and AAY, resulting in better and more enduring protection (Mazumder et al. 2023). Recent studies have also explored novel approaches for other pathogens. For instance, a multi-epitope vaccine against *S. aureus* utilized phenol-soluble modulins alpha 4 as an adjuvant (Chatterjee et al. 2021). For *K. pneumoniae*, Dar et al., demonstrated the successful affinity of an immunoinformatic-designed multi-epitope vaccine towards the inner concave interface of both TLR4 and TLR2 (Dar et al. 2019). Similarly, Elhag et al., reported that the peptides targeting Fructose Bisphosphate Aldose of *P. aeruginosa* could be recognized by B cells and T cells, thereby inducing antibody production (Elhag et al. 2020). Efforts to develop vaccines against *A. baumannii* have also shown progress. Jeffrey et al., mentioned the development of a subunit vaccine that includes whole recombinant proteins, immunogenic antigen peptides, and multi-epitopes made for discontinuous B-cell and T-cell epitopes (Jeffrey et al. 2024). Earlier studies on *M. tuberculosis* identified promising vaccine candidates, such as nitroreductase family proteins, which elicit robust immune responses due to their high antigenicity and low similarity to human proteins, showing a fight against nitrogen-associated stresses during the dormant stage of the pathogen (Shiraz et al. 2021). Various strategies are applicable in designing effective epitope-based vaccines. For instance, reverse vaccinology techniques were applied to

S. pyogenes strains to develop a multi-epitope-based sub-unit vaccine (Aslam et al. 2022; Swetha et al. 2022). These advancements underscore the potential of immunoinformatics and bioinformatics resources in revolutionizing vaccine design and enhancing protection against diverse infectious diseases.

Future perspectives

RTIs remain a major health issue at the global level in terms of morbidity and mortality. Community-acquired pathogens responsible for most cases of RTI include namely *S. pneumoniae*, *H. influenzae*, *S. aureus*, *A. baumannii*, *P. aeruginosa*, *K. pneumoniae* and *S. pyogenes* (Calderaro et al. 2022). All these pathogens have emerged as major threats, and many demonstrate MDR. The main concern about these pathogens is that they exhibit increasing antimicrobial resistance, which makes treatment more challenging and results in poor clinical outcomes. The unprecedented use of antibiotics during the COVID-19 pandemic has aggravated the AMR crisis (Stojanovic et al. 2022). Several advances have been proposed to mitigate the effects of these shortfalls through diagnostic tools for RTIs, such as syndromic and etiological diagnosis, ranging from clinical and radiological features to laboratory methods for primarily pathogen detections and host biomarkers. This has the potential to support clinical decision-making, surveillance, and follow-up of RTIs (Derbyshire and Calder 2021). Additionally, the concept of “prehabilitation”, including the role of immunonutrition, has been proposed as a novel approach to help prevent and offset RTIs (Stojanovic et al. 2022).

Overall, addressing both the current and future epidemics of RTIs and combating AMR from a ONE health approach will necessitate an individualized solution based on recent advancements in diagnostic tools, monitoring, and preventive methodologies. The success of mRNA COVID-19 vaccines has also given an incentive to innovations in vaccine technology, potentially changing the way we fight respiratory pathogens (Sheikh-Mohamed et al. 2022). The modified strategies for different populations and the development of combination vaccines could improve protection and simplify vaccination programs, especially in low-resource settings (Uchil et al. 2014). Furthermore, the applications of technologies, particularly AI, are showing promising improvement in the diagnosis and management of chronic lung diseases through personalized treatment approaches and enhanced diagnostic capabilities. Electronic health tools like telemedicine and wearables are also advancing healthcare, contributing to more precise diagnostic and better treatment monitoring in the future (Joshi et al. 2022; Honkoop et al. 2022).

Conclusion

Respiratory tract infections (RTIs) pose a significant threat to advancement of global health. There is growing evidence on the mechanism of resistance that underlies RTIs. The present review article has focused on recent advancements in understanding AMR in RTIs-causing pathogens. It encompasses various aspects of the therapeutic solutions that can be utilized to combat drug resistance. Furthermore, various computational approaches have been explored and employed to meet the AMR challenges associated with RTIs. The review provides insights into the implementation of AI and ML to combat resistance challenges through rapid and accurate AMR target identification. Employing these recent technological advances will bring down the global burden of RTIs and prevent further development of MDR pathogens, resulting in better health outcomes.

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Declarations

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References

Abavisani M, Keikha M, Karbalaee M (2024) First global report about the prevalence of multi-drug resistant *Haemophilus influenzae*:

- a systematic review and meta-analysis. *BMC Infect Dis* 24:90. <https://doi.org/10.1186/s12879-023-08930-5>
- Abayneh M, Muleta D, Simienh A et al (2022) Acute respiratory infections (ARIs) and factors associated with their poor clinical outcome among children under-five years attending pediatric wards of public hospital in Southwest district of Ethiopia: a prospective observational cohort study. *Eur J Inflam* 20:1721727X2211392. <https://doi.org/10.1177/1721727X221139266>
- Abbas R, Chakkour M, El Zein H et al (2024) General Overview of *Klebsiella pneumoniae*: epidemiology and the role of Siderophores in its pathogenicity. *Biology (Basel)* 13:78. <https://doi.org/10.3390/biology13020078>
- Abdou D (2023) Antibiotic susceptibility profiles of *Haemophilus influenzae* isolates collected in Dakar between 2018 and 2022. *Microb Sci Arch* 03:120–124. <https://doi.org/10.47587/MSA.2023.3307>
- Aceil J, Avci FY (2022) Pneumococcal surface proteins as virulence factors, immunogens, and conserved vaccine targets. *Front Cell Infect Microbiol* 12. <https://doi.org/10.3389/fcimb.2022.832254>
- Ahmadi M, Ranjbar R, Behzadi P, Mohammadian T (2022) Virulence factors, antibiotic resistance patterns, and molecular types of clinical isolates of *Klebsiella Pneumoniae*. *Expert Rev Anti Infect Ther* 20:463–472. <https://doi.org/10.1080/14787210.2022.1990040>
- Al-Bermani OK, Al-Kaim H, Ali-Malik, Saad (2020) Pathogenesis of *Streptococcus pyogenes* and immune response. *Al-Kufa Univ J Biol* 12:34–38. <https://doi.org/10.36320/ajb/v12.i3.11796>
- Alam M, Bastakoti B (2015) Therapeutic guidelines: antibiotic. Version 15. *Aust Prescr* 38:137–137. <https://doi.org/10.18773/austprescr.2015.049>
- Alamiri F, Chao Y, Baumgarten M et al (2020) A role of epithelial cells and virulence factors in Biofilm formation by *Streptococcus pyogenes* in Vitro. *Infect Immun* 88. <https://doi.org/10.1128/IAI.00133-20>
- Alamiri F, André O, De S et al (2023) Role of serotype and virulence determinants of *Streptococcus pyogenes* biofilm bacteria in internalization and persistence in epithelial cells in vitro. *Front Cell Infect Microbiol* 13. <https://doi.org/10.3389/fcimb.2023.1146431>
- AlChalabi R, Al-Rahim A, Omer D, Suleiman AA (2022) Immunoinformatics design of multi-epitope peptide-based vaccine against *Haemophilus influenzae* strain using cell division protein. *Netw Model Anal Heal Inf Bioinforma* 12:1. <https://doi.org/10.1007/s13721-022-00395-x>
- Aleem MS, Sexton R, Akella J (2024) Pneumonia in an Immunocompromised Patient
- Algammal A, Hetta HF, Mabrok M, Behzadi P (2023) Editorial: emerging multidrug-resistant bacterial pathogens superbugs: a rising public health threat. *Front Microbiol* 14. <https://doi.org/10.3389/fmicb.2023.1135614>
- Algar V, Novelli V (2007) Infections in the immunocompromised host. *Paediatr Child Health (Oxford)* 17:132–136. <https://doi.org/10.1016/j.paed.2007.02.001>
- Ali M, Reshad R, Aunkor M et al (2023) Antimicrobial Resistance: understanding the mechanism and strategies for Prevention and Control. *J Adv Biotechnol Exp Ther* 6:468. <https://doi.org/10.5455/jabet.2023.d142>
- Alsayed SSR, Gunosewoyo H (2023) Tuberculosis: Pathogenesis, current treatment regimens and new drug targets. *Int J Mol Sci* 24:5202. <https://doi.org/10.3390/ijms24065202>
- Amari S, Warda K, Bouraddane M et al (2023) Antibiotic resistance of *Streptococcus pneumoniae* in the Nasopharynx of Healthy Children Less than five Years Old after the generalization of pneumococcal vaccination in Marrakesh, Morocco. *Antibiotics* 12:442. <https://doi.org/10.3390/antibiotics12030442>
- An Z, Huang X, Zheng C, Ding W (2019) *Acinetobacter baumannii* outer membrane protein A induces HeLa cell autophagy via MAPK/JNK signaling pathway. *Int J Med Microbiol* 309:97–107. <https://doi.org/10.1016/j.ijmm.2018.12.004>
- An N, Hai L, Luong V et al (2024) Antimicrobial resistance patterns of *Staphylococcus Aureus* isolated at a General Hospital in Vietnam between 2014 and 2021. *Infect Drug Resist* Volume 17:259–273. <https://doi.org/10.2147/IDR.S437920>
- Anand U, Jacobo-Herrera N, Altemimi A, Lakhssassi N (2019) A Comprehensive Review on Medicinal plants as Antimicrobial therapeutics: potential avenues of Biocompatible Drug Discovery. *Metabolites* 9:258. <https://doi.org/10.3390/metabo9110258>
- Preethi A R, Anand Anbarasu (2023) Antimicrobial Peptides as Immunomodulators and Antimycobacterial Agents to Combat *Mycobacterium tuberculosis*: a Critical Review Probiotics and Antimicrobial Proteins 15(6):1539–1566. <https://doi.org/10.1007/s12602-022-10018-6>
- Anitha P, Anbarasu A, Ramaiah S (2016) Gene network analysis reveals the association of important functional partners involved in antibiotic resistance: a report on an important pathogenic bacterium *Staphylococcus aureus*. *Gene* 575:253–263. <https://doi.org/10.1016/j.gene.2015.08.068>
- Arif SM, Floto RA, Blundell TL (2022) Using structure-guided fragment-based drug Discovery to target *Pseudomonas aeruginosa* infections in cystic fibrosis. *Front Mol Biosci* 9. <https://doi.org/10.3389/fmolb.2022.857000>
- Ashurst JV, Dawson A (2024) *Klebsiella Pneumonia*
- Aslam S, Ashfaq UA, Zia T et al (2022) Proteome based mapping and reverse vaccinology techniques to contrive multi-epitope based subunit vaccine (MEBSV) against *Streptococcus pyogenes*. *Infect Genet Evol* 100:105259. <https://doi.org/10.1016/j.meegid.2022.105259>
- Assoni L, Girardello R, Converso TR, Darrieux M (2021) Current stage in the development of *Klebsiella pneumoniae* vaccines. *Infect Dis Ther* 10:2157–2175. <https://doi.org/10.1007/s40121-021-00533-4>
- Avendaño Carvajal L, Perret Pérez C (2020) Epidemiology of respiratory infections. *Pediatric Respiratory diseases*. Springer International Publishing, Cham, pp 263–272
- Azoulay E, Russell L, Van de Louw A et al (2020) Diagnosis of severe respiratory infections in immunocompromised patients. *Intensive Care Med* 46:298–314. <https://doi.org/10.1007/s00134-019-05906-5>
- Basu S, Varghese R, Debroy R et al (2022) Non-steroidal anti-inflammatory drugs ketorolac and etodolac can augment the treatment against pneumococcal meningitis by targeting penicillin-binding proteins. *Microb Pathog* 170:105694. <https://doi.org/10.1016/j.micpath.2022.105694>
- Basu S, Veeraraghavan B, Anbarasu A (2024) Impact of PmrB mutations on clinical *Klebsiella pneumoniae* with variable colistin-susceptibilities: structural insights and potent therapeutic solutions. *Chem Biol Drug Des* 103. <https://doi.org/10.1111/cbdd.14381>
- Baur S, Rautenberg M, Faulstich M et al (2014) A nasal epithelial receptor for *Staphylococcus aureus* WTA governs adhesion to epithelial cells and modulates nasal colonization. *PLoS Pathog* 10:e1004089. <https://doi.org/10.1371/journal.ppat.1004089>
- Behzadi P, García-Perdomo HA, Karpiński TM, Issakhanian L (2020) Metallo- β -lactamases: a review. *Mol Biol Rep* 47:6281–6294. <https://doi.org/10.1007/s11033-020-05651-9>
- Behzadi P, García-Perdomo HA, Karpiński TM (2021) Toll-like receptors: General Molecular and Structural Biology. *J Immunol Res* 2021:1–21. <https://doi.org/10.1155/2021/9914854>
- Behzadi P, Gajdacs M, Pallós P et al (2022a) Relationship between Biofilm-Formation, phenotypic virulence factors and Antibiotic Resistance in Environmental *Pseudomonas aeruginosa*. *Pathogens* 11:1015. <https://doi.org/10.3390/pathogens11091015>
- Behzadi P, Sameer AS, Nissar S et al (2022b) The Interleukin-1 (IL-1) Superfamily cytokines and their single nucleotide

- polymorphisms (SNPs). *J Immunol Res* 2022:1–25. <https://doi.org/10.1155/2022/2054431>
- Bengoechea JA, Sa Pessoa J (2019) *Klebsiella pneumoniae* infection biology: living to counteract host defences. *FEMS Microbiol Rev* 43:123–144. <https://doi.org/10.1093/femsre/fuy043>
- Berbel D, Càmarà J, González-Díaz A et al (2021) Deciphering mobile genetic elements disseminating macrolide resistance in *Streptococcus pyogenes* over a 21 year period in Barcelona, Spain. *J Antimicrob Chemother* 76:1991–2003. <https://doi.org/10.1093/jac/dkab130>
- Bethencourt Mirabal A, Nguyen AD, Ferrer G (2024) Lung Nontuberculous Mycobacterial Infections
- Bevivino A, Bacci G, Drevinek P et al (2019) Deciphering the Ecology of cystic fibrosis bacterial communities: towards systems-Level Integration. *Trends Mol Med* 25:1110–1122. <https://doi.org/10.1016/j.molmed.2019.07.008>
- Biron DG, Nedelkov D, Missé D, Holzmüller P (2011) Proteomics and host–Pathogen interactions. *Genetics and Evolution of Infectious Disease*. Elsevier, pp 263–303
- Biswas R, Swetha RG, Basu S et al (2024) Designing multi-epitope vaccine against human cytomegalovirus integrating pan-genome and reverse vaccinology pipelines. *Biologicals* 87:101782. <https://doi.org/10.1016/j.biologicals.2024.101782>
- Bouzeyen R, Javid B (2022) Therapeutic vaccines for tuberculosis: an overview. *Front Immunol* 13. <https://doi.org/10.3389/fimmu.2022.878471>
- Braverman J, Monk IR, Ge C et al (2022) *Staphylococcus aureus* specific lung resident memory CD4+Th1 cells attenuate the severity of influenza virus induced secondary bacterial pneumonia. *Mucosal Immunol* 15:783–796. <https://doi.org/10.1038/s41385-022-00529-4>
- Briles DE, Paton JC, Mukerji R et al (2019) Pneumococcal vaccines. <https://doi.org/10.1128/microbiolspec.GPP3-0028-2018>. *Microbiol Spectr* 7:
- Brooks LRK, Mias GI (2018) *Streptococcus pneumoniae*'s virulence and host immunity: Aging, Diagnostics, and Prevention. *Front Immunol* 9. <https://doi.org/10.3389/fimmu.2018.01366>
- Calderaro A, Buttrini M, Farina B et al (2022) Respiratory Tract Infections and Laboratory Diagnostic methods: a review with a focus on Syndromic Panel-based assays. *Microorganisms* 10:1856. <https://doi.org/10.3390/microorganisms10091856>
- Černiauskienė K, Dambrauskienė A, Vitkauskienė A (2023) Associations between β -Lactamase types of *Acinetobacter baumannii* and Antimicrobial Resistance. *Med (B Aires)* 59:1386. <https://doi.org/10.3390/medicina59081386>
- Chakravarty S, Massé E (2019) RNA-Dependent regulation of virulence in pathogenic Bacteria. *Front Cell Infect Microbiol* 9. <https://doi.org/10.3389/fcimb.2019.00337>
- Chandy S, Manoharan A, Hameed A et al (2022) A study on pediatric respiratory tract infections in hospitalised children from Chennai. *Clin Epidemiol Glob Heal* 15:101067. <https://doi.org/10.1016/j.cegh.2022.101067>
- Chang A, Adlowitz DG, Yellamattay E, Pichichero M (2010) Haemophilus influenzae outer membrane protein P6 molecular characterization may not differentiate all strains of H. Influenzae from H. Haemolyticus. *J Clin Microbiol* 48:3756–3757. <https://doi.org/10.1128/JCM.01255-10>
- Chang RYK, Wallin M, Lin Y et al (2018) Phage therapy for respiratory infections. *Adv Drug Deliv Rev* 133:76–86. <https://doi.org/10.1016/j.addr.2018.08.001>
- Chatterjee R, Sahoo P, Mahapatra SR et al (2021) Development of a conserved chimeric vaccine for induction of strong Immune response against *Staphylococcus aureus* using Immunoinformatics approaches. *Vaccines* 9:1038. <https://doi.org/10.3390/vaccines9091038>
- Chen C-L, Dudek A, Liang Y-H et al (2022) d-mannose-sensitive pilus of *Acinetobacter baumannii* is linked to biofilm formation and adherence onto respiratory tract epithelial cells. *J Microbiol Immunol Infect* 55:69–79. <https://doi.org/10.1016/j.jmii.2021.01.008>
- Childs A, Zullo AR, Joyce NR et al (2019) The burden of respiratory infections among older adults in long-term care: a systematic review. *BMC Geriatr* 19:210. <https://doi.org/10.1186/s12877-019-1236-6>
- Cillóniz C, García-Vidal C, Ceccato A, Torres A (2018) Antimicrobial Resistance among *Streptococcus pneumoniae*. *Antimicrobial Resistance in the 21st Century*. Springer International Publishing, Cham, pp 13–38
- Conn GL, Bavro VN, Davies C (2019) Editorial: bacterial mechanisms of Antibiotic Resistance: a structural perspective. *Front Mol Biosci* 6. <https://doi.org/10.3389/fmolb.2019.00071>
- Connor NE, Islam MS, Mullany LC et al (2022) Risk factors for community-acquired bacterial infection among young infants in South Asia: a longitudinal cohort study with nested case–control analysis. *BMJ Glob Heal* 7:e009706. <https://doi.org/10.1136/bmjgh-2022-009706>
- Costain G, Cohn RD, Scherer SW, Marshall CR (2021) Genome sequencing as a diagnostic test. *Can Med Assoc J* 193:E1626–E1629. <https://doi.org/10.1503/cmaj.210549>
- Curran CS, Bolig T, Torabi-Parizi P (2018) Mechanisms and targeted therapies for *Pseudomonas aeruginosa* Lung infection. *Am J Respir Crit Care Med* 197:708–727. <https://doi.org/10.1164/rccm.201705-1043SO>
- Dar HA, Zaheer T, Shehroz M et al (2019) Immunoinformatics-aided design and evaluation of a potential Multi-epitope Vaccine against *Klebsiella pneumoniae*. *Vaccines* 7:88. <https://doi.org/10.3390/vaccines7030088>
- de Miguel-Díez J, López-de-Andrés A, Hernández-Barrera V et al (2017) Impact of COPD on outcomes in hospitalized patients with community-acquired pneumonia: analysis of the Spanish national hospital discharge database (2004–2013). *Eur J Intern Med* 43:69–76. <https://doi.org/10.1016/j.ejim.2017.06.008>
- Deinhardt-Emmer S, Haupt KF, García-Moreno M et al (2019) *Staphylococcus aureus* Pneumonia: Preceding Influenza infection paves the way for low-virulent strains. *Toxins (Basel)* 11:734. <https://doi.org/10.3390/toxins11120734>
- Deng X, den Bakker HC, Hendriksen RS (2016) Genomic epidemiology: whole-genome-sequencing-powered surveillance and Outbreak Investigation of Foodborne bacterial pathogens. *Annu Rev Food Sci Technol* 7:353–374. <https://doi.org/10.1146/annurev-food-041715-033259>
- Derbyshire EJ, Calder PC (2021) Respiratory tract infections and antibiotic resistance: a protective role for vitamin D? <https://doi.org/10.3389/fnut.2021.652469>. *Front Nutr* 8:
- Dhillon BK, Smith M, Baghela A et al (2020) Systems Biology approaches to understanding the human Immune System. <https://doi.org/10.3389/fimmu.2020.01683>. *Front Immunol* 11:
- Dion CF, Ashurst JV (2024) *Streptococcus pneumoniae*
- Dörries K, Schlueter R, Lalk M (2014) Impact of antibiotics with various Target sites on the Metabolome of *Staphylococcus aureus*. *Antimicrob Agents Chemother* 58:7151–7163. <https://doi.org/10.1128/AAC.03104-14>
- Douradinha B (2024) Exploring the journey: a comprehensive review of vaccine development against *Klebsiella pneumoniae*. *Microbiol Res* 287:127837. <https://doi.org/10.1016/j.micres.2024.127837>
- Du D, Wang-Kan X, Neuberger A et al (2018) Multidrug efflux pumps: structure, function and regulation. *Nat Rev Microbiol* 16:523–539. <https://doi.org/10.1038/s41579-018-0048-6>
- Du S, Vilhena C, King S et al (2021) Molecular analyses identifies new domains and structural differences among *Streptococcus*

- pneumoniae immune evasion proteins PspC and hic. *Sci Rep* 11:1701. <https://doi.org/10.1038/s41598-020-79362-3>
- Edgeworth JD (2023) Respiratory metagenomics: route to routine service. *Curr Opin Infect Dis* 36:115–123. <https://doi.org/10.1097/QCO.0000000000000909>
- Ekinici E, Willen L, Rodriguez Ruiz JP et al (2023) Haemophilus influenzae carriage and antibiotic resistance profile in Belgian infants over a three-year period (2016–2018). <https://doi.org/10.3389/fmicb.2023.1160073>. *Front Microbiol* 14:
- El Ashkar S, Osman M, Rafei R et al (2017) Molecular detection of genes responsible for macrolide resistance among Streptococcus pneumoniae isolated in North Lebanon. *J Infect Public Health* 10:745–748. <https://doi.org/10.1016/j.jiph.2016.11.014>
- Elhag M, Alaagib RM, Ahmed NM et al (2020) Design of Epitope-based peptide vaccine against Pseudomonas aeruginosa Fructose Bisphosphate aldolase protein using Immunoinformatics. *J Immunol Res* 2020:1–11. <https://doi.org/10.1155/2020/9475058>
- Ellington MJ, Ekelund O, Aarestrup FM et al (2017) The role of whole genome sequencing in antimicrobial susceptibility testing of bacteria: report from the EUCAST Subcommittee. *Clin Microbiol Infect* 23:2–22. <https://doi.org/10.1016/j.cmi.2016.11.012>
- Fan X, Liu X, Ji L et al (2018) Epidemiological analysis and rapid detection by one-step multiplex PCR assay of Haemophilus influenzae in children with respiratory tract infections in Zhejiang Province, China. *BMC Infect Dis* 18:414. <https://doi.org/10.1186/s12879-018-3295-2>
- Fan J, Toth I, Stephenson RJ (2024) Recent scientific advancements towards a vaccine against Group A Streptococcus. *Vaccines* 12:272. <https://doi.org/10.3390/vaccines12030272>
- Fischetti VA (2019) Vaccine approaches to protect against Group A Streptococcal Pharyngitis. *Microbiol Spectr* 7. <https://doi.org/10.1128/microbiolspec.GPP3-0010-2018>
- Fontaine MC, Lee JJ, Kehoe MA (2003) Combined contributions of streptolysin O and streptolysin S to virulence of serotype M5 Streptococcus pyogenes strain Manfredo. *Infect Immun* 71:3857–3865. <https://doi.org/10.1128/IAI.71.7.3857-3865.2003>
- Fothergill JL, Neill DR, Loman N et al (2014) Pseudomonas aeruginosa adaptation in the nasopharyngeal reservoir leads to migration and persistence in the lungs. *Nat Commun* 5:4780. <https://doi.org/10.1038/ncomms5780>
- Freschi L, Vincent AT, Jeukens J et al (2019) The Pseudomonas aeruginosa Pan-genome provides New insights on its Population structure, horizontal gene transfer, and Pathogenicity. *Genome Biol Evol* 11:109–120. <https://doi.org/10.1093/gbe/evy259>
- Gao NJ, Uchiyama S, Pill L et al (2021) Site-specific conjugation of Cell Wall Polyrhamnose to protein SpyAD envisioning a safe Universal Group A Streptococcal Vaccine. *Infect Microbes Dis* 3:87–100. <https://doi.org/10.1097/IM9.0000000000000044>
- Garcia-Clemente M, de la Rosa D, Máiz L et al (2020) Impact of Pseudomonas aeruginosa infection on patients with chronic inflammatory Airway diseases. *J Clin Med* 9. <https://doi.org/10.3390/jcm9123800>
- GBD 2016 Lower Respiratory Infections Collaborators (2018) Estimates of the global, regional, and national morbidity, mortality, and aetiologies of lower respiratory infections in 195 countries, 1990–2016: a systematic analysis for the global burden of Disease Study 2016. *Lancet Infect Dis* 18:1191–1210. [https://doi.org/10.1016/S1473-3099\(18\)30310-4](https://doi.org/10.1016/S1473-3099(18)30310-4)
- Geisinger E, Huo W, Hernandez-Bird J, Isberg RR (2019) Acinetobacter baumannii: envelope determinants that control Drug Resistance, Virulence, and Surface Variability. *Annu Rev Microbiol* 73:481–506. <https://doi.org/10.1146/annurev-micro-020518-115714>
- Golshahi L, Seed KD, Dennis JJ, Finlay WH (2008) Toward Modern Inhalational bacteriophage therapy: nebulization of bacteriophages of Burkholderia cepacia Complex. *J Aerosol Med Pulm Drug Deliv* 21:351–360. <https://doi.org/10.1089/jamp.2008.0701>
- Grad YH, Lipsitch M (2014) Epidemiologic data and pathogen genome sequences: a powerful synergy for public health. *Genome Biol* 15:538. <https://doi.org/10.1186/s13059-014-0538-4>
- Grzela K, Zagórska W, Grzela T (2012) Mechanisms of the innate immunity in the respiratory system. *Cent Eur J Immunol* 3:280–285. <https://doi.org/10.5114/ceji.2012.30807>
- Guirado E, Amat I, Gil O et al (2006) Passive serum therapy with polyclonal antibodies against Mycobacterium tuberculosis protects against post-chemotherapy relapse of tuberculosis infection in SCID mice. *Microbes Infect* 8:1252–1259. <https://doi.org/10.1016/j.micinf.2005.12.004>
- Guiton AK, Wright GD (2018) Antimicrobial Resistance and respiratory infections. *Chest* 154:1202–1212. <https://doi.org/10.1016/j.chest.2018.06.019>
- Hancock REW, Nijnik A, Philpott DJ (2012) Modulating immunity as a therapy for bacterial infections. *Nat Rev Microbiol* 10:243–254. <https://doi.org/10.1038/nrmicro2745>
- Hassanzadeh H, Baber J, Begier E et al (2023) Efficacy of a 4-Antigen Staphylococcus aureus Vaccine in spinal surgery: the STaphylococcus aureus suRgical Inpatient Vaccine Efficacy (STRIVE) Randomized Clinical Trial. *Clin Infect Dis* 77:312–320. <https://doi.org/10.1093/cid/ciad218>
- Hawser S, Lociuo S, Islam K (2006) Dihydrofolate reductase inhibitors as antibacterial agents. *Biochem Pharmacol* 71:941–948. <https://doi.org/10.1016/j.bcp.2005.10.052>
- He Y, Liu WJ, Jia N et al (2023) Viral respiratory infections in a rapidly changing climate: the need to prepare for the next pandemic. *eBioMedicine* 93:104593. <https://doi.org/10.1016/j.ebiom.2023.104593>
- Hibbert TM, Whiteley M, Renshaw SA et al (2023) Emerging strategies to target virulence in Pseudomonas aeruginosa respiratory infections. *Crit Rev Microbiol* 1–16. <https://doi.org/10.1080/1040841X.2023.2285995>
- Hilleringmann M, Giusti F, Baudner BC et al (2008) Pneumococcal pili are composed of protofilaments exposing Adhesive clusters of Rrg A. *PLoS Pathog* 4:e1000026. <https://doi.org/10.1371/journal.ppat.1000026>
- Holler JG, Christensen SB, Slotved H-C et al (2012) Novel inhibitory activity of the Staphylococcus aureus NorA efflux pump by a kaempferol rhamnoside isolated from Persea lingue nees. *J Antimicrob Chemother* 67:1138–1144. <https://doi.org/10.1093/jac/dks005>
- Honkoop P, Usmani O, Bonini M (2022) The current and future role of Technology in Respiratory Care. *Pulm Ther* 8:167–179. <https://doi.org/10.1007/s41030-022-00191-y>
- Huy TXN (2024) Overcoming Klebsiella pneumoniae antibiotic resistance: new insights into mechanisms and drug discovery. *Beni-Suef Univ J Basic Appl Sci* 13:13. <https://doi.org/10.1186/s43088-024-00470-4>
- Inoue K, Kinoshita M, Muranishi K et al (2023) Effect of a Novel Trivalent Vaccine Formulation against Acute Lung Injury caused by Pseudomonas aeruginosa. *Vaccines* 11:1088. <https://doi.org/10.3390/vaccines11061088>
- Jafari D, Malih S, Gomari MM et al (2020) Designing a chimeric subunit vaccine for influenza virus, based on HA2, M2e and CTxB: a bioinformatics study. *BMC Mol Cell Biol* 21:89. <https://doi.org/10.1186/s12860-020-00334-6>
- Jain S, Self WH, Wunderink RG et al (2015) Community-Acquired Pneumonia requiring hospitalization among U.S. adults. *N Engl J Med* 373:415–427. <https://doi.org/10.1056/NEJMoa1500245>
- Jain R, Beckett VV, Konstan MW et al (2018) KB001-A, a novel anti-inflammatory, found to be safe and well-tolerated in cystic fibrosis patients infected with Pseudomonas aeruginosa. *J Cyst Fibros* 17:484–491. <https://doi.org/10.1016/j.jcf.2017.12.006>

- Jedrzejak MJ (2001) Pneumococcal virulence factors: structure and function. *Microbiol Mol Biol Rev* 65:187–207. <https://doi.org/10.1128/MMBR.65.2.187-207.2001>
- Jeffreys S, Tompkins MP, Aki J et al (2024) Development and evaluation of an Immunoinformatics-based Multi-peptide Vaccine against *Acinetobacter baumannii* infection. *Vaccines* 12:358. <https://doi.org/10.3390/vaccines12040358>
- Jensen A, Valdórrson O, Frimodt-Møller N et al (2015) Commensal *Streptococci* serve as a Reservoir for β -Lactam resistance genes in *Streptococcus pneumoniae*. *Antimicrob Agents Chemother* 59:3529–3540. <https://doi.org/10.1128/AAC.00429-15>
- Johnson MM, Odell JA (2014) Nontuberculous mycobacterial pulmonary infections. *J Thorac Dis* 6:210–220. <https://doi.org/10.3978/j.issn.2072-1439.2013.12.24>
- Joshi T, Sharma P, Joshi T et al (2022) Repurposing of FDA approved drugs against *Salmonella enteric* serovar Typhi by targeting dihydrofolate reductase: an in silico study. *J Biomol Struct Dyn* 40:3731–3744. <https://doi.org/10.1080/07391102.2020.1850356>
- Joshi T, Vijayakumar S, Ghosh S et al (2024) Identifying Novel therapeutics for the resistant mutant F533L in PBP3 of *Pseudomonas aeruginosa* using ML techniques. *ACS Omega* 9:28046–28060. <https://doi.org/10.1021/acsomega.4c00929>
- Kaczmarek FS, Gootz TD, Dib-Hajj F et al (2004) Genetic and molecular characterization of β -Lactamase-negative ampicillin-resistant *Haemophilus influenzae* with unusually high resistance to Ampicillin. *Antimicrob Agents Chemother* 48:1630–1639. <https://doi.org/10.1128/AAC.48.5.1630-1639.2004>
- Kadioglu A, Weiser JN, Paton JC, Andrew PW (2008) The role of *Streptococcus pneumoniae* virulence factors in host respiratory colonization and disease. *Nat Rev Microbiol* 6:288–301. <https://doi.org/10.1038/nrmicro1871>
- Kapoor G, Saigal S, Elongavan A (2017) Action and resistance mechanisms of antibiotics: a guide for clinicians. *J Anaesthesiol Clin Pharmacol* 33:300–305. https://doi.org/10.4103/joacp.JOACP_349_15
- Karampatakis T, Tsergouli K, Behzadi P (2023) Carbapenem-resistant *Klebsiella pneumoniae*: virulence factors, molecular epidemiology and latest updates in Treatment options. *Antibiotics* 12:234. <https://doi.org/10.3390/antibiotics12020234>
- Karampatakis T, Tsergouli K, Behzadi P (2024) Pan-genome plasticity and virulence factors: a natural treasure trove for *Acinetobacter baumannii*. *Antibiotics* 13:257. <https://doi.org/10.3390/antibiotics13030257>
- Karlsson R, Thorsell A, Gomila M et al (2020) Discovery of species-unique peptide biomarkers of bacterial pathogens by Tandem Mass Spectrometry-based Proteotyping. *Mol Cell Proteom* 19:518–528. <https://doi.org/10.1074/mcp.RA119.001667>
- Kebede D, Admas A, Mekonnen D (2021) Prevalence and antibiotics susceptibility profiles of *Streptococcus pyogenes* among pediatric patients with acute pharyngitis at Felege Hiwot Comprehensive Specialized Hospital, Northwest Ethiopia. *BMC Microbiol* 21:135. <https://doi.org/10.1186/s12866-021-02196-0>
- Khare T, Anand U, Dey A et al (2021) Exploring phytochemicals for combating Antibiotic Resistance in Microbial pathogens. *Front Pharmacol* 12. <https://doi.org/10.3389/fphar.2021.720726>
- Khattak ZE, Anjum F (2023) *Haemophilus influenzae* Infection
- Killough M, Rodgers A, Ingram R (2022) *Pseudomonas aeruginosa*: recent advances in Vaccine Development. *Vaccines* 10:1100. <https://doi.org/10.3390/vaccines10071100>
- Kim G-L, Seon S-H, Rhee D-K (2017) Pneumonia and *Streptococcus pneumoniae* vaccine. *Arch Pharm Res* 40:885–893. <https://doi.org/10.1007/s12272-017-0933-y>
- King P (2012) *Haemophilus influenzae* and the lung (*Haemophilus* and the lung). *Clin Transl Med*. <https://doi.org/10.1186/2001-1326-1-10.1>
- Kongkham B, Prabakaran D, Puttaswamy H (2020) Opportunities and challenges in managing antibiotic resistance in bacteria using plant secondary metabolites. *Fitoterapia* 147:104762. <https://doi.org/10.1016/j.fitote.2020.104762>
- Kyriakidis I, Vasileiou E, Pana ZD, Tragiannidis A (2021) *Acinetobacter baumannii* Antibiotic Resistance mechanisms. *Pathogens* 10:373. <https://doi.org/10.3390/pathogens10030373>
- Lange P (2009) Chronic obstructive pulmonary disease and risk of infection. *Pneumonol Alergol Pol* 77:284–288
- Lazar H, Horn MP, Zuercher AW et al (2009) Pharmacokinetics and Safety Profile of the human Anti- *Pseudomonas aeruginosa* Serotype O11 Immunoglobulin M monoclonal antibody KBPA-101 in healthy volunteers. *Antimicrob Agents Chemother* 53:3442–3446. <https://doi.org/10.1128/AAC.01699-08>
- LeClaire RD, Hunt RE, Bavari S (2002) Protection against Bacterial Superantigen Staphylococcal Enterotoxin B by Passive Vaccination. *Infect Immun* 70:2278–2281. <https://doi.org/10.1128/IAI.70.5.2278-2281.2002>
- Letourneau AR, Issa NC, Baden LR (2014) Pneumonia in the immunocompromised host. *Curr Opin Pulm Med* 20:272–279. <https://doi.org/10.1097/MCP.0000000000000051>
- Li C-X, Gao J, Zhang Z et al (2022) Multiomics integration-based molecular characterizations of COVID-19. <https://doi.org/10.1093/bib/bbab485>. *Brief Bioinform* 23:
- Li P, Zhang S, Wang J et al (2023) Uncovering the Secretion systems of *Acinetobacter baumannii*: structures and functions in pathogenicity and antibiotic resistance. *Antibiotics* 12:195. <https://doi.org/10.3390/antibiotics12020195>
- Liang M, Fan Y, Zhang D et al (2022) Metagenomic next-generation sequencing for accurate diagnosis and management of lower respiratory tract infections. *Int J Infect Dis* 122:921–929. <https://doi.org/10.1016/j.ijid.2022.07.060>
- Liu GY (2009) Molecular pathogenesis of *Staphylococcus aureus* infection. *Pediatr Res* 65:71. <https://doi.org/10.1203/PDR.0b013e31819dc44d>. R-77R
- Liu K, Yang W, Dong X et al (2016) Inhalation study of *Mycobacteriophage D29* Aerosol for mice by Endotracheal Route and nose-only exposure. *J Aerosol Med Pulm Drug Deliv* 29:393–405. <https://doi.org/10.1089/jamp.2015.1233>
- Lotfi M, Hamblin MR, Rezaei N (2020) COVID-19: transmission, prevention, and potential therapeutic opportunities. *Clin Chim Acta* 508:254–266. <https://doi.org/10.1016/j.cca.2020.05.044>
- Loughran AJ, Orihuela CJ, Tuomanen EI (2019) *Streptococcus pneumoniae*: Invasion and inflammation. *Microbiol Spectr* 7. <https://doi.org/10.1128/microbiolspec.GPP3-0004-2018>
- Luthra S, Rominski A, Sander P (2018) The role of antibiotic-target-modifying and antibiotic-modifying enzymes in *Mycobacterium abscessus* Drug Resistance. *Front Microbiol* 9. <https://doi.org/10.3389/fmicb.2018.02179>
- Manohar P, Loh B, Athira S, et al (2020) Secondary Bacterial Infections During Pulmonary Viral Disease: Phage Therapeutics as Alternatives to Antibiotics? *Front Microbiol* 11. <https://doi.org/10.3389/fmicb.2020.01434>
- Manoharan A, Manchanda V, Balasubramanian S et al (2017) Invasive pneumococcal disease in children aged younger than 5 years in India: a surveillance study. *Lancet Infect Dis* 17:305–312. [https://doi.org/10.1016/S1473-3099\(16\)30466-2](https://doi.org/10.1016/S1473-3099(16)30466-2)
- Martin LW, Robson CL, Watts AM et al (2018) Expression of *Pseudomonas aeruginosa* Antibiotic Resistance genes varies greatly during infections in cystic fibrosis patients. *Antimicrob Agents Chemother* 62. <https://doi.org/10.1128/AAC.01789-18>
- Matesanz López C, Loras Gallego C, Cacho Calvo J et al (2021) Patients with non-tuberculous mycobacteria in respiratory samples: a 5-year epidemiological study. *Rev Esp Quimioter* 34:120–125. <https://doi.org/10.37201/req/121.2020>

- Mazumder L, Shahab M, Islam S et al (2023) An immunoinformatics approach to epitope-based vaccine design against PspA in *Streptococcus pneumoniae*. *J Genet Eng Biotechnol* 21:57. <https://doi.org/10.1186/s43141-023-00506-9>
- Mekuria S, Seyoum A, Ataro Z et al (2022) Prevalence, Antimicrobial Resistance, and Associated factors of *Streptococcus pneumoniae* colonization rate among old-age patients with respiratory tract infection attending Sheik Hassan Yebere Referral and Karamara General Hospitals, Jigjiga, Ethiopia. *Can J Infect Dis Med Microbiol* 2022:1–9. <https://doi.org/10.1155/2022/9338251>
- Miller RR, Montoya V, Gardy JL et al (2013) Metagenomics for pathogen detection in public health. *Genome Med* 5:81. <https://doi.org/10.1186/gm485>
- Miriti DM, Muthini JM, Nyamache AK (2023) Study of bacterial respiratory infections and antimicrobial susceptibility profile among antibiotics naive outpatients visiting Meru teaching and referral hospital, Meru County, Kenya in 2018. *BMC Microbiol* 23:172. <https://doi.org/10.1186/s12866-023-02905-x>
- Miryala SK, Basu S, Naha A et al (2021) Identification of bioactive natural compounds as efficient inhibitors against *Mycobacterium tuberculosis* protein-targets: a molecular docking and molecular dynamics simulation study. *J Mol Liq* 341:117340. <https://doi.org/10.1016/j.molliq.2021.117340>
- Morris FC, Dexter C, Kostoulas X et al (2019) The mechanisms of Disease caused by *Acinetobacter baumannii*. *Front Microbiol* 10. <https://doi.org/10.3389/fmicb.2019.01601>
- Mücke P-A, Maaß S, Kohler TP et al (2020) Proteomic adaptation of *Streptococcus pneumoniae* to the human antimicrobial peptide LL-37. *Microorganisms* 8:413. <https://doi.org/10.3390/microorganisms8030413>
- Naito Y, Moriyama K, Sawa T (2017) Anti-PcrV Immunization for *Pseudomonas aeruginosa* Pneumonia in Cystic Fibrosis. In: Progress in Understanding Cystic Fibrosis. InTech
- Nakahashi-Ouchida R, Mori H, Yuki Y et al (2022) Induction of mucosal IgA-Mediated protective immunity against nontypeable *Haemophilus influenzae* infection by a Cationic Nanogel-based P6 nasal vaccine. *Front Immunol* 13. <https://doi.org/10.3389/fimmu.2022.819859>
- Nandhini P, Kumar P, Mickymaray S et al (2022) Recent developments in Methicillin-Resistant *Staphylococcus aureus* (MRSA) treatment: a review. *Antibiotics* 11:606. <https://doi.org/10.3390/antibiotics11050606>
- Netea MG, Azam T, Lewis EC et al (2006) *Mycobacterium tuberculosis* induces interleukin-32 production through a caspase-1/IL-18/interferon-gamma-dependent mechanism. *PLoS Med* 3:e277. <https://doi.org/10.1371/journal.pmed.0030277>
- Novotny L, Jurcisek J, Godfroid F et al (2006) Passive immunization with human anti-protein D antibodies induced by polysaccharide protein D conjugates protects chinchillas against otitis media after intranasal challenge with *Haemophilus influenzae*. *Vaccine* 24:4804–4811. <https://doi.org/10.1016/j.vaccine.2006.03.021>
- Okafor CN, Rewane A, Momodu II (2024) *Bacillus Calmette Guerin*
- Oliveira D, Borges A, Simões M (2018) *Staphylococcus aureus* Toxins and their molecular activity in Infectious diseases. *Toxins (Basel)* 10:252. <https://doi.org/10.3390/toxins10060252>
- Pabary R, Singh C, Morales S et al (2016) Antipseudomonal Bacteriophage reduces infective Burden and Inflammatory Response in Murine Lung. *Antimicrob Agents Chemother* 60:744–751. <https://doi.org/10.1128/AAC.01426-15>
- Panditrao G, Bhowmick R, Meena C, Sarkar RR (2022) Emerging landscape of molecular interaction networks: opportunities, challenges and prospects. *J Biosci* 47:24. <https://doi.org/10.1007/s12038-022-00253-y>
- Paonessa JR, Shah RD, Pickens CI et al (2019) Rapid Detection of Methicillin-Resistant *Staphylococcus aureus* in BAL. *Chest* 155:999–1007. <https://doi.org/10.1016/j.chest.2019.02.007>
- Patarčić I, Gelemanović A, Kirin M et al (2015) The role of host genetic factors in respiratory tract infectious diseases: systematic review, meta-analyses and field synopsis. *Sci Rep* 5:16119. <https://doi.org/10.1038/srep16119>
- Peek SF, Ollivett TL, Divers TJ (2018) Respiratory diseases. *Rebhun's diseases of dairy cattle*. Elsevier, pp 94–167
- Peela SCM, Basu S, Sharma J et al (2023) Structure elucidation and Interaction Dynamics of MefA-MsrD Efflux Proteins in *Streptococcus pneumoniae*: impact on Macrolide susceptibility. *ACS Omega* 8:39454–39467. <https://doi.org/10.1021/acsomega.3c05210>
- Peltola V, Shih S-R, To KK-W (2020) Editorial: respiratory virus infection: recent advances. *Front Med* 7. <https://doi.org/10.3389/fmed.2020.00257>
- Pivard M, Moreau K, Vandenesch F (2021) *Staphylococcus aureus* Arsenal to conquer the lower respiratory tract. <https://doi.org/10.1128/mSphere.00059-21>. *mSphere* 6:
- Poole K (2007) Efflux pumps as antimicrobial resistance mechanisms. *Ann Med* 39:162–176. <https://doi.org/10.1080/07853890701195262>
- Popoff MR (2024) Overview of bacterial protein toxins from pathogenic Bacteria: Mode of Action and insights into Evolution. *Toxins (Basel)* 16:182. <https://doi.org/10.3390/toxins16040182>
- Prazak J, Iten M, Cameron DR et al (2019) Bacteriophages improve outcomes in experimental *Staphylococcus aureus* Ventilator-associated Pneumonia. *Am J Respir Crit Care Med* 200:1126–1133. <https://doi.org/10.1164/rccm.201812-2372OC>
- Priyamvada P, Swetha RG, Dasgupta R et al (2024) Immunoinformatics strategy for designing a multi-epitope chimeric vaccine to combat *Neisseria gonorrhoeae*. *Vacunas* 25:385–402. <https://doi.org/10.1016/j.vacun.2024.04.003>
- Pulzova L, Navratilova L, Comor L (2017) Alterations in outer membrane permeability Favor Drug-Resistant phenotype of *Klebsiella pneumoniae*. *Microb Drug Resist* 23:413–420. <https://doi.org/10.1089/mdr.2016.0017>
- Qin S, Xiao W, Zhou C et al (2022) *Pseudomonas aeruginosa*: pathogenesis, virulence factors, antibiotic resistance, interaction with host, technology advances and emerging therapeutics. *Signal Transduct Target Ther* 7:199. <https://doi.org/10.1038/s41392-022-01056-1>
- Ragunathan A, Malathi K, Ramaiah S, Anbarasu A (2019) FtsA as a cidal target for *Staphylococcus aureus*: molecular docking and dynamics studies. *J Cell Biochem* 120:7751–7758. <https://doi.org/10.1002/jcb.28049>
- Ranjbar A, Rasooli I, Jahangiri A, Ramezanalizadeh F (2022) Specific egg yolk antibody raised to biofilm associated protein (bap) is protective against murine pneumonia caused by *Acinetobacter baumannii*. *Sci Rep* 12:12576. <https://doi.org/10.1038/s41598-022-16894-w>
- Ratnatunga CN, Lutzky VP, Kupz A et al (2020) The rise of Non-tuberculosis Mycobacterial Lung Disease. <https://doi.org/10.3389/fimmu.2020.00303>. *Front Immunol* 11:
- Ravi M, Krishnasamy L, Krishnakumar S (2023) Antimicrobial susceptibility pattern of bacterial isolates in patients with lower respiratory tract infections. *Int J Life Sci Pharma Res*. <https://doi.org/10.22376/ijlpr.2023.13.5.L371-L376>
- Reygaert WC (2018) An overview of the antimicrobial resistance mechanisms of bacteria. *AIMS Microbiol* 4:482–501. <https://doi.org/10.3934/microbiol.2018.3.482>
- Reynolds D, Kollef M (2021) The epidemiology and Pathogenesis and treatment of *Pseudomonas aeruginosa* infections: an update. *Drugs* 81:2117–2131. <https://doi.org/10.1007/s40265-021-01635-6>
- Rostock L, Driller R, Grätz S et al (2018) Molecular insights into antibiotic resistance - how a binding protein traps albicidin. *Nat Commun* 9:3095. <https://doi.org/10.1038/s41467-018-05551-4>

- Roy A, Swetha RG, Basu S et al (2024) Integrating pan-genome and reverse vaccinology to design multi-epitope vaccine against herpes simplex virus type-1. *3 Biotech* 14:176. <https://doi.org/10.1007/s13205-024-04022-6>
- Saleri N, Ryan ET (2019) Respiratory infections. *Travel Medicine*. Elsevier, pp 527–537
- Sandoval MM, Ruvinsky S, Palermo MC et al (2024) Antimicrobial resistance of *Streptococcus pneumoniae* from invasive pneumococcal diseases in latin American countries: a systematic review and meta-analysis. *Front Public Heal* 12. <https://doi.org/10.3389/fpubh.2024.1337276>
- Sarshar M, Behzadi P, Scribano D et al (2021) *Acinetobacter baumannii*: an ancient commensal with weapons of a Pathogen. *Pathogens* 10:387. <https://doi.org/10.3390/pathogens10040387>
- Sastre-Femenia MA, Fernández-Muñoz A, Gomis-Font MA et al (2023) *Pseudomonas aeruginosa* antibiotic susceptibility profiles, genomic epidemiology and resistance mechanisms: a nation-wide five-year time lapse analysis. *Lancet Reg Heal - Eur* 34:100736. <https://doi.org/10.1016/j.lanepe.2023.100736>
- Sattar SBA, Sharma S (2024) Bacterial Pneumonia
- Scafa-Udriste A, Popa M-I, Popa G-L (2023) Updates on Staphylococcal vaccines. *Microbiol Res (Pavia)* 15:137–151. <https://doi.org/10.3390/microbiolres15010009>
- Sethi S (2010) Infection as a comorbidity of COPD. *Eur Respir J* 35:1209–1215. <https://doi.org/10.1183/09031936.00081409>
- Sharma A, Ahmad Farouk I, Lal SK (2021) COVID-19: a review on the Novel Coronavirus Disease Evolution, Transmission, Detection, Control and Prevention. *Viruses* 13. <https://doi.org/10.3390/v13020202>
- Sheikh-Mohamed S, Isho B, Chao GYC et al (2022) Systemic and mucosal IgA responses are variably induced in response to SARS-CoV-2 mRNA vaccination and are associated with protection against subsequent infection. *Mucosal Immunol* 15:799–808. <https://doi.org/10.1038/s41385-022-00511-0>
- Shi J, Cheng J, Liu S et al (2024) *Acinetobacter baumannii*: an evolving and cunning opponent. *Front Microbiol* 15. <https://doi.org/10.3389/fmicb.2024.1332108>
- Shiraz M, Lata S, Kumar P et al (2021) Immunoinformatics analysis of antigenic epitopes and designing of a multi-epitope peptide vaccine from putative nitro-reductases of *Mycobacterium tuberculosis* DosR. *Infect Genet Evol* 94:105017. <https://doi.org/10.1016/j.meegid.2021.105017>
- Shriram V, Khare T, Bhagwat R et al (2018) Inhibiting bacterial drug efflux pumps via Phyto-Therapeutics to combat threatening Antimicrobial Resistance. *Front Microbiol* 9. <https://doi.org/10.3389/fmicb.2018.02990>
- Shrivastava S, Shrivastava P, Ramasamy J (2018) World health organization releases global priority list of antibiotic-resistant bacteria to guide research, discovery, and development of new antibiotics. *J Med Soc* 32:76. https://doi.org/10.4103/jms.jms_25_17
- Singh R, Capalash N, Sharma P (2017) Immunoprotective potential of BamA, the outer membrane protein assembly factor, against MDR *Acinetobacter baumannii*. *Sci Rep* 7:12411. <https://doi.org/10.1038/s41598-017-12789-3>
- Singh R, Capalash N, Sharma P (2022) Vaccine development to control the rising scourge of antibiotic-resistant *Acinetobacter baumannii*: a systematic review. *3 Biotech* 12:85. <https://doi.org/10.1007/s13205-022-03148-9>
- Slack MPE, Cripps AW, Grimwood K et al (2021) Invasive *Haemophilus influenzae* infections after 3 decades of hib protein Conjugate Vaccine Use. *Clin Microbiol Rev* 34:e0002821. <https://doi.org/10.1128/CMR.00028-21>
- Stojanovic Z, Gonçalves-Carvalho F, Marín A et al (2022) Advances in diagnostic tools for respiratory tract infections: from tuberculosis to COVID-19 – changing paradigms? *ERJ Open Res* 8:00113–02022. <https://doi.org/10.1183/23120541.00113-2022>
- Subramanian K, Henriques-Normark B, Normark S (2019) Emerging concepts in the pathogenesis of the *Streptococcus pneumoniae*: from nasopharyngeal colonizer to intracellular pathogen. *Cell Microbiol* 21. <https://doi.org/10.1111/cmi.13077>
- Sudhakar P, Machiels K, Verstockt B et al (2021) Computational Biology and Machine Learning Approaches to understand mechanistic microbiome-host interactions. *Front Microbiol* 12. <https://doi.org/10.3389/fmicb.2021.618856>
- Swetha RG, Basu S, Ramaiah S, Anbarasu A (2022) Multi-epitope Vaccine for Monkeypox using Pan-genome and Reverse Vaccinology approaches. *Viruses* 14:2504. <https://doi.org/10.3390/v14112504>
- Tam K, Torres VJ (2019) *Staphylococcus aureus* secreted toxins and extracellular enzymes. *Microbiol Spectr* 7. <https://doi.org/10.1128/microbiolspec.GPP3-0039-2018>
- Teng F, Xiong X, Zhang S et al (2022) Efficacy Assessment of Phage Therapy in treating *Staphylococcus aureus*-Induced mastitis in mice. *Viruses* 14:620. <https://doi.org/10.3390/v14030620>
- Tong SYC, Davis JS, Eichenberger E et al (2015) *Staphylococcus aureus* infections: Epidemiology, Pathophysiology, Clinical manifestations, and management. *Clin Microbiol Rev* 28:603–661. <https://doi.org/10.1128/CMR.00134-14>
- Trinh TD, Zasowski EJ, Claeys KC et al (2017) Multidrug-resistant *Pseudomonas aeruginosa* lower respiratory tract infections in the intensive care unit: prevalence and risk factors. *Diagn Microbiol Infect Dis* 89:61–66. <https://doi.org/10.1016/j.diagmicrobio.2017.06.009>
- Tristram S, Jacobs MR, Appelbaum PC (2007) Antimicrobial Resistance in *Haemophilus influenzae*. *Clin Microbiol Rev* 20:368–389. <https://doi.org/10.1128/CMR.00040-06>
- Tsang RSW, Ulanova M (2017) The changing epidemiology of invasive *Haemophilus influenzae* disease: emergence and global presence of serotype a strains that may require a new vaccine for control. *Vaccine* 35:4270–4275. <https://doi.org/10.1016/j.vaccine.2017.06.001>
- Tuon FF, Dantas LR, Suss PH, Tasca Ribeiro VS (2022) Pathogenesis of the *Pseudomonas aeruginosa* Biofilm: a review. *Pathogens* 11:300. <https://doi.org/10.3390/pathogens11030300>
- Uppalapati SR, Sett A, Pathania R (2020) The outer membrane proteins OmpA, CarO, and OprD of *Acinetobacter baumannii* Confer a two-Pronged Defense in Facilitating its success as a Potent Human Pathogen. <https://doi.org/10.3389/fmicb.2020.589234>. *Front Microbiol* 11:
- Uchil RR, Kohli GS, Katekhaye VM, Swami OC (2014) Strategies to combat antimicrobial resistance. *J Clin Diagn Res* 8:ME01-4. <https://doi.org/10.7860/JCDR/2014/8925.4529>
- Varghese R, Basu S, Neeravi A et al (2022) Emergence of Meropenem Resistance among Cefotaxime non-susceptible *Streptococcus pneumoniae*: evidence and challenges. *Front Microbiol* 12. <https://doi.org/10.3389/fmicb.2021.810414>
- Vidor E, Hoffenbach A, Fletcher MA (2001) *Haemophilus influenzae* type b vaccine: reconstitution of lyophilised PRP-T vaccine with a pertussis-containing paediatric combination vaccine, or a change in the primary Series Immunisation schedule, May modify the serum Anti-PRP antibody responses. *Curr Med Res Opin* 17:197–209. <https://doi.org/10.1185/0300799039117063>
- Vijayakumar S, Kumar H, Basu S et al (2024a) Changing Landscape of Antimicrobial Resistance in neonatal Sepsis: an in silico analyses of Multidrug Resistance in *Klebsiella pneumoniae*. *Pediatr Infect Dis J*. <https://doi.org/10.1097/INF.0000000000004358>
- Vijayakumar S, Swetha RG, Bakthavatchalam YD et al (2024b) Genomic investigation unveils colistin resistance mechanism in carbapenem-resistant *Acinetobacter baumannii* clinical isolates. *Microbiol Spectr* 12. <https://doi.org/10.1128/spectrum.02511-23>
- Vollset SE, Ababneh HS, Abate YH et al (2024) Burden of disease scenarios for 204 countries and territories, 2022–2050: a forecasting

- analysis for the global burden of Disease Study 2021. *Lancet* 403:2204–2256. [https://doi.org/10.1016/S0140-6736\(24\)00685-8](https://doi.org/10.1016/S0140-6736(24)00685-8)
- Wachino J, Jin W, Kimura K, Arakawa Y (2019) Intercellular transfer of chromosomal antimicrobial resistance genes between *Acinetobacter baumannii* strains mediated by Prophages. *Antimicrob Agents Chemother* 63. <https://doi.org/10.1128/AAC.00334-19>
- Wang Q, Liu C-G, Xu J et al (2021) Characteristics and antibiotic resistance of *Haemophilus influenzae* in children with Lower Respiratory Tract Infection in Chengdu, China. *Jundishapur J Microbiol* 14. <https://doi.org/10.5812/jjm.114210>
- Weidmann MD, Berry GJ, Green DA, Wu F (2022) Prevalence and clinical disease severity of respiratory coinfections during the Coronavirus Disease 2019 Pandemic. *Adv Mol Pathol* 5:73–84. <https://doi.org/10.1016/j.yamp.2022.07.003>
- Yang N, Jin X, Zhu C et al (2023) Subunit vaccines for *Acinetobacter baumannii*. *Front Immunol* 13. <https://doi.org/10.3389/fimmu.2022.1088130>
- Zeng X, Gu H, Peng L et al (2020) Transcriptome profiling of Lung Innate Immune responses potentially Associated with the pathogenesis of *Acinetobacter baumannii* Acute Lethal Pneumonia. <https://doi.org/10.3389/fimmu.2020.00708>. *Front Immunol* 11:
- Zhanell GG, Chung P, Adam H et al (2014) Ceftolozane/Tazobactam: a novel Cephalosporin/ β -Lactamase inhibitor combination with activity against Multidrug-Resistant Gram-Negative Bacilli. *Drugs* 74:31–51. <https://doi.org/10.1007/s40265-013-0168-2>
- Zhang F, Cheng W (2022) The mechanism of bacterial resistance and potential bacteriostatic strategies. *Antibiotics* 11:1215. <https://doi.org/10.3390/antibiotics11091215>
- Zhang B-Z, Hu D, Dou Y et al (2021) Identification and Evaluation of Recombinant Outer Membrane Proteins as vaccine candidates against *Klebsiella pneumoniae*. *Front Immunol* 12. <https://doi.org/10.3389/fimmu.2021.730116>
- Zhang W, Zhou H, Jiang Y et al (2022) *Acinetobacter baumannii* outer membrane protein A induces pulmonary epithelial barrier dysfunction and bacterial translocation through the TLR2/IQGAP1 Axis. *Front Immunol* 13. <https://doi.org/10.3389/fimmu.2022.927955>
- Zuercher AW, Horn MP, Que JU et al (2006) Antibody responses induced by long-term vaccination with an octovalent conjugate *Pseudomonas aeruginosa* vaccine in children with cystic fibrosis. *FEMS Immunol Med Microbiol* 47:302–308. <https://doi.org/10.1111/j.1574-695X.2006.00103.x>

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